

Particulate Matter

Week in Review

Week ending 22 August 2026 | 16 August 2026 to 22 August 2026 | 112 records pooled from 7 daily issues

112

RECORDS POOLED
(112 UNIQUE DOIs)

52%

ON THE MEASURE-
MENT
SIDE (58 RECORDS)

10

SUBTOPICS
ACTIVE

41

POOLED EFFECT
ESTIMATES

22

ABSTRACT-FREE
(20% OF WEEK)

The week in one paragraph

Six independent groups stopped analysing $PM_{2.5}$ as a mass this week and started analysing it as a list of constituents — and in the two endpoints where a single constituent led, it was the same one. Meanwhile a fifth of the week's corpus arrived with no abstract, which is the price this pipeline is now paying for its own coverage.

The methodological through-line is constituent decomposition, and it is unusually consistent. **Wu et al.** (16 Aug) put prenatal $PM_{2.5}$ and its components through a mixture model for childhood overweight; **Wang et al.** (16 Aug) do the same for early-life anaemia across China, India and 26 African countries; **Zhao et al.** (17 Aug) report spontaneous abortion by constituent, with **ammonium highest (OR 1.19) — above total mass (1.18)**; **Parmar et al.** (18 Aug) separate black carbon from bulk $PM_{2.5}$ for neuropsychiatric hospitalisation in Delhi and find the component term nearly an order of magnitude larger; **Chen et al.** (21 Aug) run five constituents against incident depression in 24,940 UK Biobank participants and again land on *ammonium*; and **Wei et al.** (21 Aug) run all fifteen components against mortality in 528,127 lung-cancer patients by weighted quantile sum.

The answers do not agree, and that is the finding. Ammonium and nitrate — the secondary inorganic fraction — lead in the reproductive and psychiatric endpoints, while Wei's mortality weights fall on silicon, vanadium, zinc and iron, a crustal, traffic and heavy-fuel-oil signature. Two different exposures are being called $PM_{2.5}$. The week also supplies the atmospheric-chemistry half of that argument: **Chai et al.** (21 Aug) measure secondary inorganic aerosol at **49.55% of $PM_{2.5}$ mass** in suburban Nanjing and show the system is total-nitrate-limited, so the very fraction that the epidemiology is now singling out is the one that does *not* respond to ammonia control. If a component drives the risk, mass-only monitoring cannot see it and mass-targeted policy cannot move it.

Against that, an operational fact that should be read as a cost. 22 of 112 records (20%) carry no abstract from any checked source and are classified but unread — almost double last week's 15 of 134 (11%). *Metadata only* is now the single largest architecture label in the pooled corpus, ahead of every real study design. This follows directly from widening the Crossref by-ISSN sweep: the leg buys coverage of journals PubMed indexes late or not at all, and pays for it in publishers that deposit a title and a DOI and nothing more.

Subtopic trend and inflection

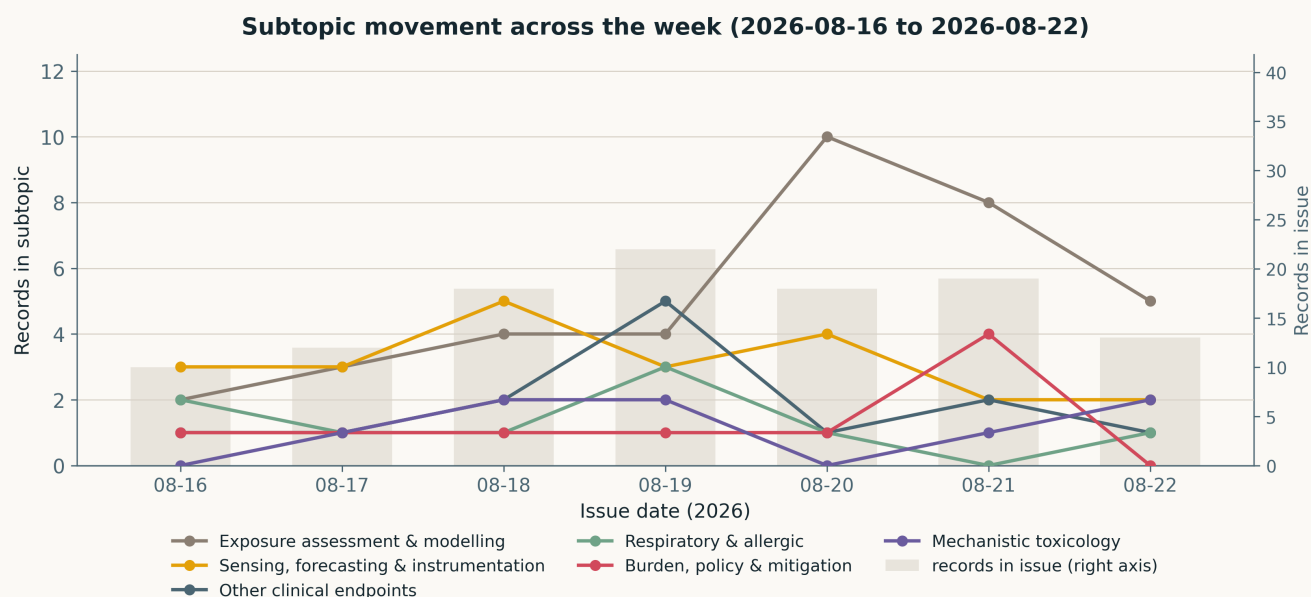


Fig. Per-issue subtopic counts across the week from `state/metrics.csv`. **Read the denominators first:** daily totals were 10, 12, 18, 22, 18, 19 and 13 — a range of 2.2×, and a flatter week than 9–15 August (12 to 28). The 16 August floor of 10 is a Sunday and the smallest issue the watch has recorded; the 22 August figure of 13 is a Saturday. **No day in this week carries a publisher batch artefact of the kind that distorted 11–12 August**, so unlike last week the trend line can be read more or less at face value.

The measurement side took the week back. *Exposure assessment & modelling* rose 26 → 36 while *Sensing, forecasting & instrumentation* fell 34 → 22; together they are 58 of 112 records, a measurement-side share of 52% against last week's 45% and 54% the week before. **Do not read this as a recovery in instrumentation.** The sensing cluster shrank; what grew is exposure and deposition work, and four of the eight *Exposure* records in the 21 August issue alone measure PM by what it lands on — pine needles, privet leaves, tree canopies — rather than by what it does in air. The oscillation between 54%, 45% and 52% across three weeks is within the range this pipeline's own harvesting legs can produce, and no claim about the field should be built on it.

Contracting, and worth watching. *Occupational & indoor* fell 9 → 4, *Mechanistic toxicology* 11 → 8 and *Cardiovascular & metabolic* 7 → 3. The mechanistic drop breaks a two-week run at exactly 11 and is the first movement in that cluster since the watch began keeping the count. Cardiovascular at 3 is the lowest recorded for what is normally the field's largest health cluster — against 10 respiratory-endpoint records this week. One week is not a trend; two would be.

Holding. *Other clinical endpoints* rose 11 → 13, extending last week's finding that the clinical tail is now longer than the cardiorespiratory trunk, and the content stays unfamiliar: cirrhotic hepatic decompensation, diabetic microvascular complications, systemic lupus, and — new this week — the prognosis of an emerging tick-borne viral infection. All ten subtopics were active on at least one day.

Study architecture

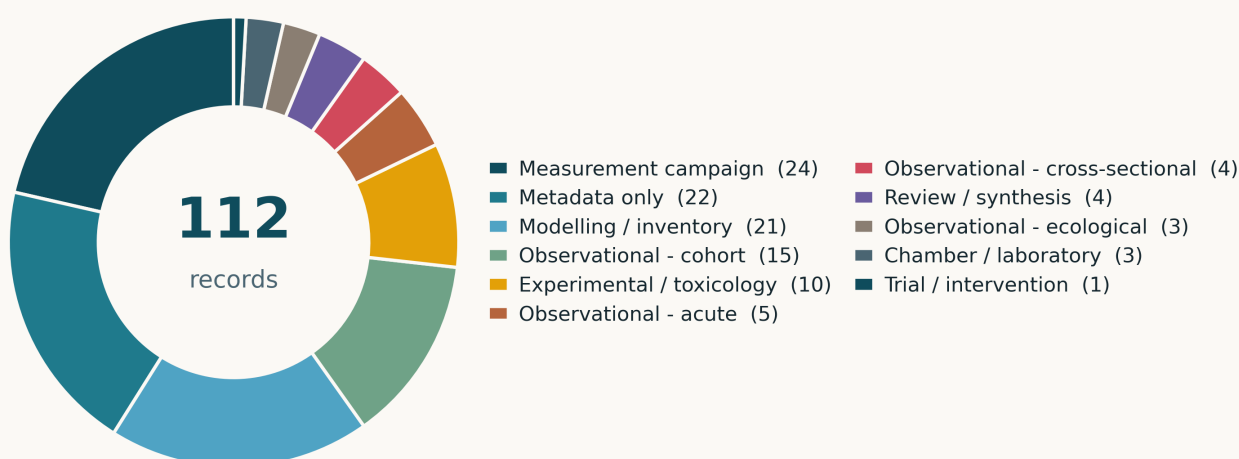


Fig. Pooled study architecture over 112 records. **The largest slice is a defect, not a design.** *Metadata only* is 22 records, 20% of the week — surfaced with no abstract in Crossref or Europe PMC and therefore classified from title and journal alone — and it now outranks measurement campaigns, cohorts and every other architecture in the corpus. It concentrates on two days (six on 19 August, six on 22 August) and on four Elsevier titles. Against that, *Observational — cohort* at 15 and *Experimental / toxicology* at 9 are the load-bearing evidence classes. *Trial / intervention* is 1 record in 112: a single filtration-intervention effectiveness study on 19 August. For the fourth week running, no completed randomised intervention on a PM exposure entered the corpus.

Pooled analytics

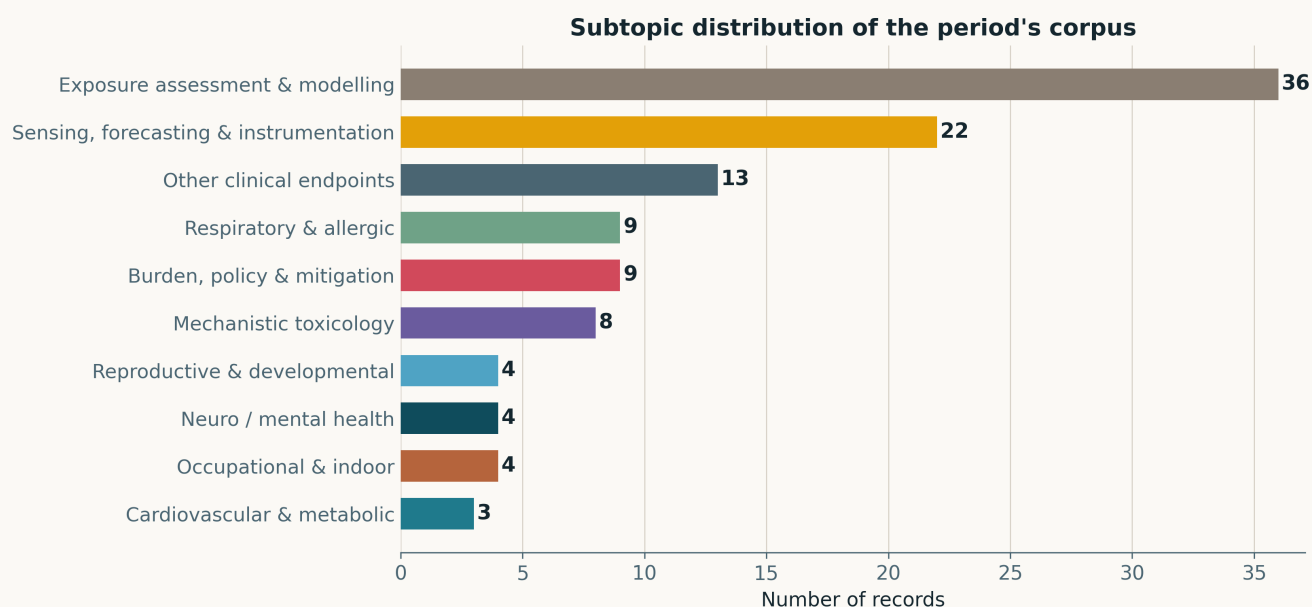


Fig. Pooled subtopic distribution, 16–22 August. *Exposure assessment & modelling* (36) and *Sensing* (22) together are just over half the week; *Other clinical endpoints* (13) is the largest health cluster, ahead of *Respiratory & allergic* (9) and *Burden, policy & mitigation* (9). Every one of the ten canonical clusters is occupied, which has now held for four consecutive weeks.

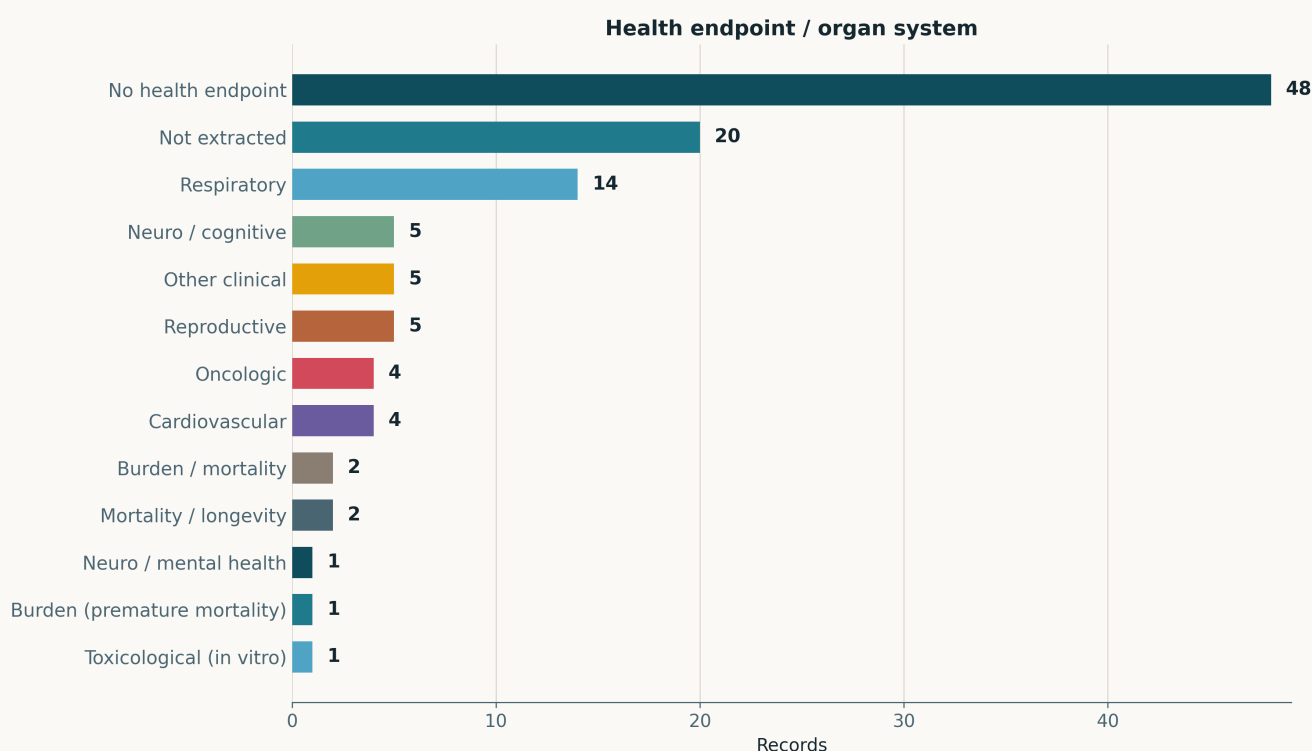


Fig. Health endpoint by organ system across the pooled week. 62 of 112 records carry no human health endpoint at all — 20 because no abstract was deposited and the endpoint could not be extracted, and the remaining 42 because they are monitoring, exposure, instrument or aerosol-chemistry work by design. Of what is left, respiratory endpoints lead at 10, with a further 4 specifically respiratory *infection*; neuro/cognitive and cardiovascular follow.

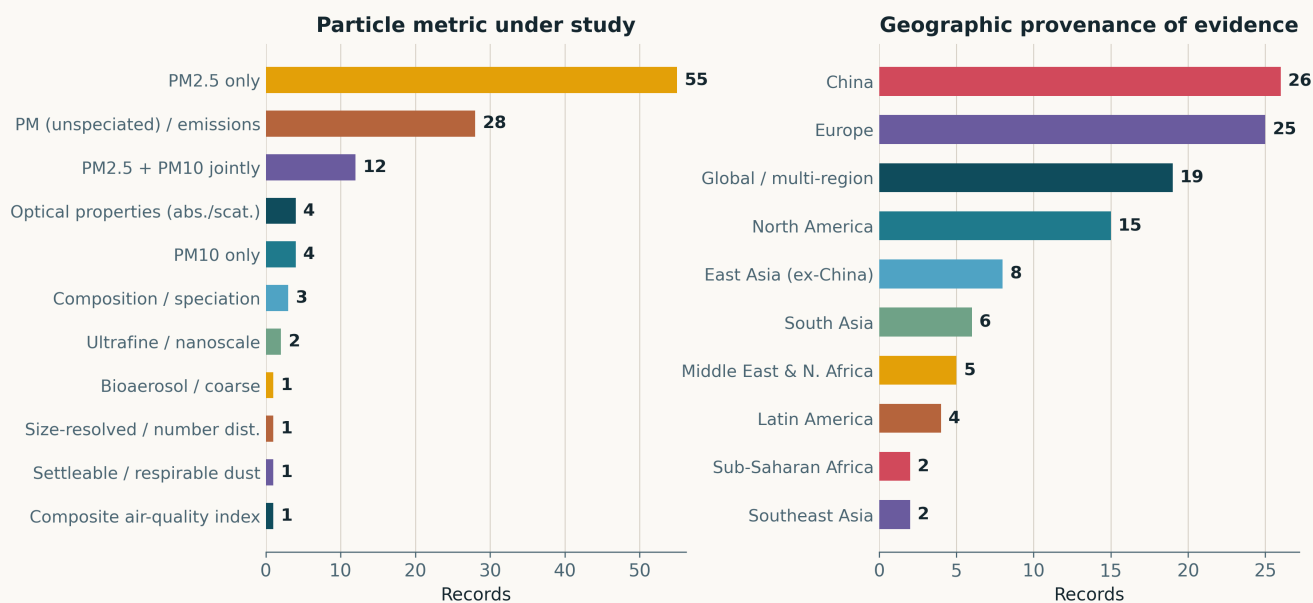
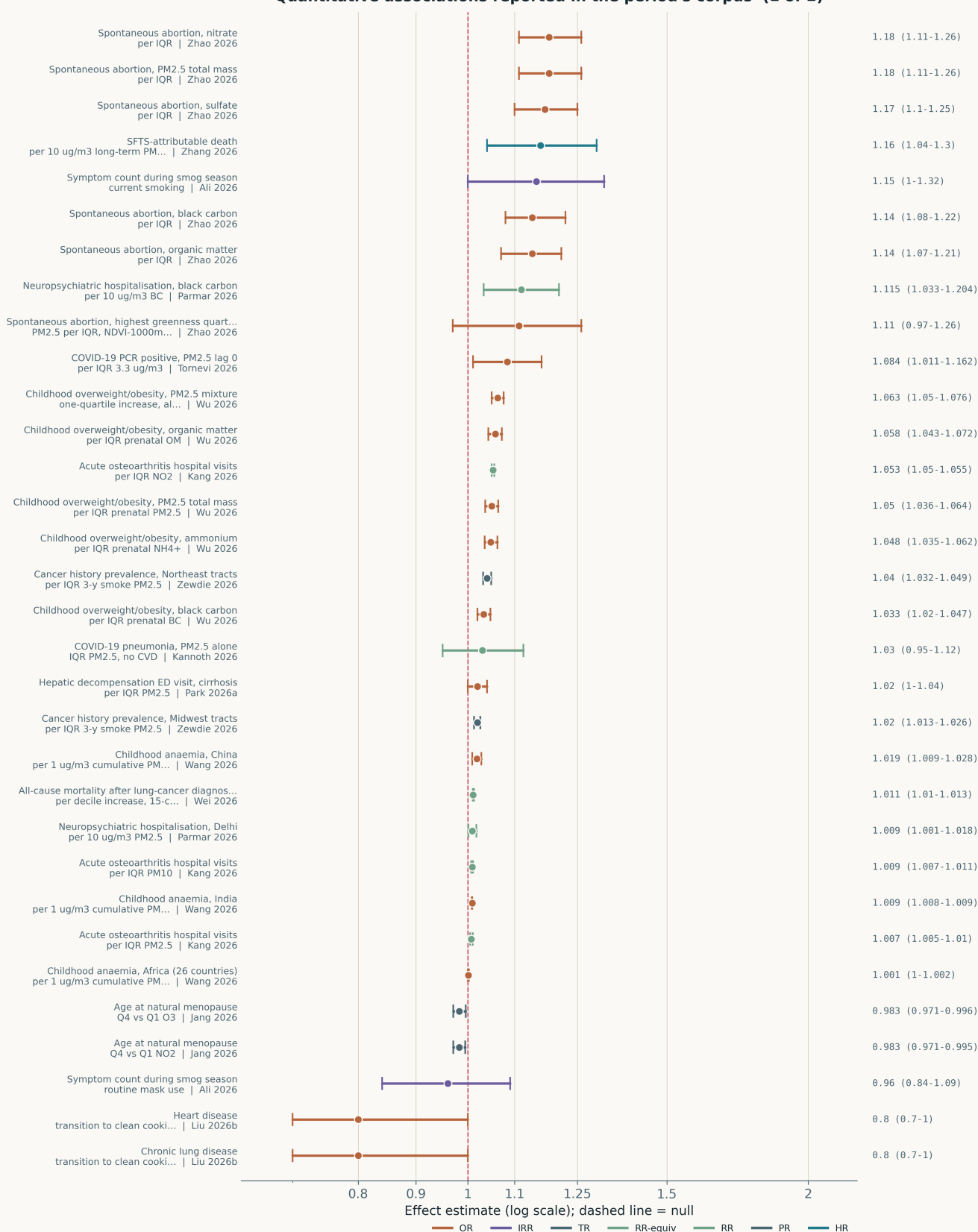


Fig. Left — particle metric across the pooled week. PM_{2.5} alone is the exposure in 55 records and appears jointly with PM₁₀ in 12; **ultrafine particles are the explicit subject of 2 records in 112**, and size-resolved number distributions of one more. That gap has been stable for the entire watch and is the clearest standing mismatch between what is measured and what deposition physics says matters. Right — geographic provenance. **Europe (25) came within one record of China (26)**, which has not happened before in this watch; North America is 15. Sub-Saharan Africa and Southeast Asia contribute two records each.

Quantitative associations reported in the period's corpus (1 of 2)



Quantitative associations reported in the period's corpus (2 of 2)

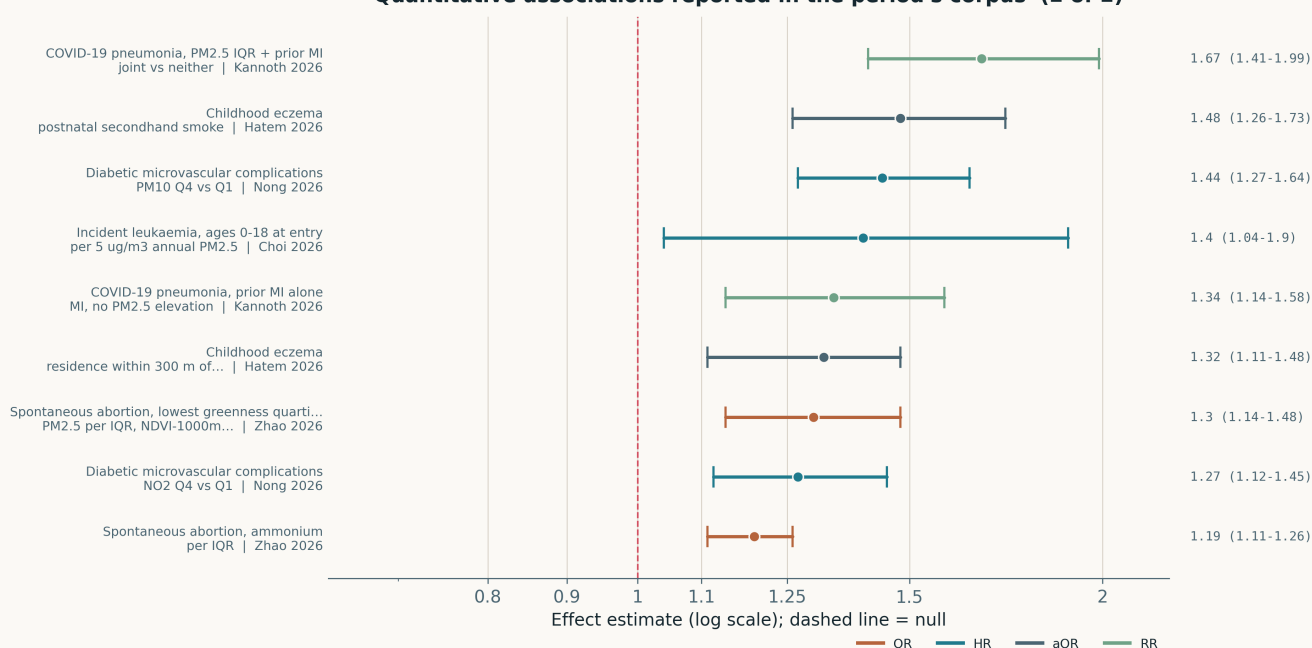


Fig. All 41 pooled effect estimates, split across two panels at 32 rows each and plotted on a common logarithmic axis. **Exposure contrasts are heterogeneous and the estimates are not mutually comparable** — per-IQR, per-10 $\mu\text{g}/\text{m}^3$, per-SD, quartile contrasts and binary residential exposures all appear — so read direction and precision, never relative magnitude, and **no meta-analytic summary is computed or implied**. Two structural features are worth naming. First, **the week is front-loaded**: 30 of the 41 estimates come from 16–18 August, 20 August produced none at all, and 21 August produced one. Second, **the constituent-resolved studies cluster tightly just above the null** — Wu's five childhood-obesity ORs span 1.03–1.06 and Zhao's seven spontaneous-abortion ORs span 1.14–1.19 — because a per-unit constituent contrast is a much smaller exposure step than a quartile or threshold contrast, not because the effects are weaker. The two estimates left of the null are both from the same paper: the CHARLS clean-fuel transition.

Life-course windows addressed this period (records may span several stages)

Preconception / in utero **Infancy & early childhood** **Puberty & adolescence** **Working-age adulthood** **Older adults (65+)**



Prenatal PM2.5 constituted mixtures; prenatal MERRA-2 window, Spontaneous abortion from residential cohort entered at age 0, London/Luton school cohort case-crossover; UK Biobank ages into 65+ over 13.5 y of AF follow-up, Korean OA visits skew older;

Fig. Pooled life-course coverage. Per-stage counts and annotations are summed and merged across the seven daily issues rather than inherited from any one of them. The in-utero and early-childhood windows carry the week, consistent with the developmental mixture cohorts (Wu, Wang, Zhao, Perera) that also supply most of the constituent-resolved estimates above.

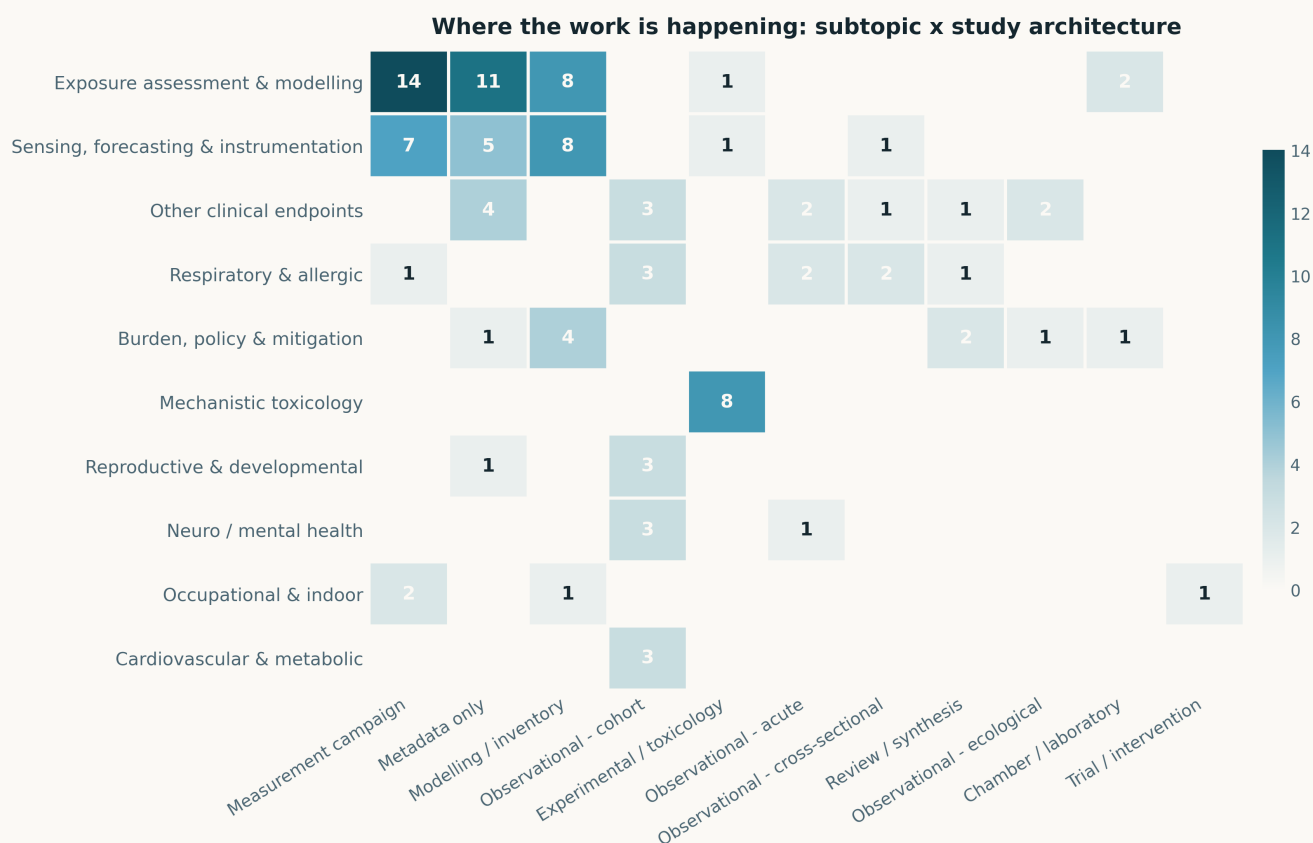


Fig. Subtopic × architecture over the pooled week. The *Metadata only* column spreads across most subtopic rows rather than sitting in one corner — the abstract-free deposits are not a single topic's problem. **The structural gap the watch has reported every week is visible again:** the *Sensing* and *Exposure* rows contain almost no cells that also carry a health endpoint, and the cohort column contains almost no cells that measured their own exposure. Instrumentation and epidemiology remain methodologically separate populations of paper.

Venues, provenance and what the pipeline did

Venues. *Environ Sci Technol* led with 11 records, then *Atmos Chem Phys* (9), *Atmos Environ* (8) and *Environ Pollut* (6); *Ecotoxicol Environ Saf*, *Atmos Pollut Res*, *Atmosphere* and *Atmos Res* contributed 4 each. **Five preprints** entered the week (Research Square 3, medRxiv 1, Preprints.org 1) — 4.5% of the corpus, and three of the five carried quantitative results substantial enough to lead a daily issue.

Two harvester changes were made inside this period, and both affect what the figures above show. On 22 August the supplementary ISSN list trialled on 20 August was promoted into the standing Crossref sweep — 15 additions, each ISSN verified against Crossref's own journal index, with MDPI *Sensors* deliberately excluded after returning 27 deposits for zero in-scope records. The immediate effect was that the Crossref leg carried 62% of the 22 August issue while PubMed returned two records. The immediate cost is the abstract-free share discussed above. **The first run with *Environ Sci Technol* in the list also returned 154 works of which 142 were mastheads and issue-information stubs** being re-deposited in bulk from volumes 42–52; those carry DOIs and titles and would have passed every downstream check, so a title-pattern front-matter filter now drops them at the source.

Three figure defects were closed during the period and one is disclosed uncorrected. The forest-plot axis now enforces a minimum $1.6\times$ span in log space, after a single near-null estimate on 21 August was auto-scaled onto an axis spanning 1.5% and drawn two-thirds of the way to the right edge. Two design labels (*In vitro* exposure, *In vitro* organ-on-chip) and one more (*Health impact model*) were silently falling into "Other / mixed" on every issue and rollout that contained them, and *Literature review* was mapped to *Tool / software*; all four are corrected, and this rollout is the first output rendered with the corrections. **Disclosed and not corrected:** the endpoint-label canonicalisation gap reported in the 15 August rollout persists, so f2b's "no health endpoint" bar is assembled here by hand in the caption (62 of 112) rather than being read off the figure.

OpenAlex has now returned HTTP 429 on twenty consecutive runs and should be treated as removed from the pipeline rather than degraded. Europe PMC's `CREATION_DATE` leg continues to work but is increasingly dominated by archival back-scans — legacy PMC page images with no title and a publication year decades outside the window — which accounted for 14 of 28 unique Europe PMC hits on 21 August and 15 of 23 on 22 August. **The trial registry has not been re-queried since 8 August**; rollups do not query APIs, and the standing trial watch is therefore stale by two weeks and is not restated in this issue.

Open gaps

- ▶ **Nobody measured the exposure they analysed.** Of the 20 tier-A human health records this week, the overwhelming majority inherit a modelled exposure surface — satellite, land-use regression, or a spatiotemporal reanalysis product — and none reports that model's error at the spatial scale of its own analysis. The single clear exception is Liu et al. (22 Aug), whose exposure is a directly observed household fuel transition. This is now the fourth consecutive week the watch has recorded this and it remains the largest methodological opening for an instrumentation group.
- ▶ **Constituent-resolved epidemiology has arrived without constituent-resolved monitoring.** Six papers this week analyse $PM_{2.5}$ by component, and all six take those components from chemical transport or satellite-derived products. No record this week measured a speciated ambient time series and linked it to a health outcome in the same study.
- ▶ **Ultrafine particles: 2 records in 112.** Unchanged in direction and magnitude from every previous week of this watch.
- ▶ **One intervention study in 112 records, and it was not randomised.** The evidence base being built is almost entirely observational, while the policy questions the same papers pose — which precursor to cut, which fuel to subsidise, which filter to deploy — are interventional.
- ▶ **Twenty per cent of the week is unread.** 22 records have no abstract from any checked source. Two of them (aerosol pH and preterm birth; ammonia-driven ROS generation in secondary organic aerosol) sit directly on the week's main thread and are the highest priority for full-text retrieval before the monthly rollout.

Paper-level digest — the week by subtopic

Every record that appeared in a daily issue keeps that issue's critical summary verbatim; nothing here is re-written or re-assessed. Records surfaced without an abstract are shown as such. Ordered by subtopic, then by the date of the issue that first carried the record.

Sensing, forecasting & instrumentation

22 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (22 OF 22 RECORDS IN THIS SUBTOPIC)

Ayele et al. 2026 Environ Sci Atmos

Fine-scale black carbon and TSP composition at a heavy-traffic intersection, Addis Ababa

METADATA ONLY — the deposited abstract is truncated to its opening sentences and reports no quantities. What is verifiable from it: the study addresses spatiotemporal variability of equivalent black carbon and the molecular composition of organics in total suspended particulate at a roadside intersection, and states that Ethiopian urban aerosol remains poorly characterised. Today's only sub-Saharan record.

doi: 10.1039/d6ea00100a

Band & Mohammadi 2026 (preprint) arXiv

Shift-aware dual-encoder transfer learning for short-horizon PM_{2.5} forecasting

A source encoder pretrained on hourly PM_{2.5} from 10 US monitoring locations is fused with a target-specific encoder and adapted to **77 Taiwanese stations** over two years under a chronological train/validation/test split — the correct protocol, and one many comparable papers do not use. Against four baselines the frozen-source dual encoder gives **MSE 21.8960, MAE 3.1597, $R^2 = 0.8725$** (a 7.1% and 4.1% MSE reduction over two transfer-learning variants); allowing the source encoder to adapt improves this to **MSE 21.6575, MAE 3.1383, $R^2 = 0.8739$** . Ablation shows the Taiwan-specific branch matters most, and SHAP attributes predictions mainly to recent PM_{2.5} and meteorology. The framing — explicit handling of source/target distribution shift to avoid negative transfer — is the right one for anyone standing up a network in a region with no long record. Two reservations: the absolute gain over the better baselines is a few percent of MSE and no significance or repeated-seed variance is reported, and both domains are reference-grade regulatory data, so transfer onto low-cost nodes (a far larger shift) is untested. Preprint, not peer reviewed.

doi: 10.48550/arXiv.2608.14456

Clark & Bashar 2026 Risk Anal

How IoT enables the Protective Action Decision Model during a wildfire event

Survey of 1,200 Oregon residents (Jun–Aug 2021) about their responses to the 2020 fire season, framed against the Protective Action Decision Model and the availability of hyper-local real-time PM_{2.5} from IoT devices. The finding that matters for network designers is negative: residents were **overwhelmed by the data and reported inadequate guidance** on how to act on it. Retrospective self-report a year after the event, and the survey does not establish who actually had sensor access — so this constrains interface and messaging design rather than measuring sensor efficacy. Still the rare paper in this watch that treats the last mile of a dense network as the problem.

doi: 10.1111/risa.70319

Dal Porto et al. 2026 Aerosol Sci Technol

Humidified CAPS-PM_{SSA} with a revised truncation-correction methodology

Metadata only. Taylor & Francis withheld the abstract on this run and Semantic Scholar was rate-limited throughout, so nothing below the title, authors and journal is verified and no result is asserted. On the registered title alone the paper describes a humidified variant of the cavity-attenuated phase-shift single-scattering-albedo monitor together with an improved method for the scattering truncation correction — the two quantities that dominate uncertainty when an SSA measurement is used as a reference for hygroscopic growth. Relevant enough to the humidity-correction problem to carry rather than defer; flagged for abstract retrieval on the next run.

doi: 10.1080/02786826.2026.2711663

Diana et al. 2026 Talanta

Validated multi-technique workflow for inorganic characterisation of brake-wear PM₁₀

Thirty-two PM₁₀ samples from WLTP-based brake-dynamometer tests on four tribological couples — low-steel (LS) and non-asbestos organic (NAO) pads against grey cast-iron (GCI) or coated discs. A combined acid digestion (HNO₃/H₂O₂ plus HNO₃/HF/H₃BO₃) was validated against **NIST SRM 1648a** and quantified by ICP-OES and ICP-MS, giving **27 elements with recoveries above 75% for most analytes**; FESEM-EDS confirmed the trends at particle scale and PCA/HCA separated the tribological families cleanly. **LS-GCI produced the highest PM₁₀ emission factors and NAO-GCI the lowest**; Fe dominates LS-derived material while Ba, Ti, K and Zr mark NAO. An Mn-based enrichment factor improved discrimination of rotor- versus pad-derived contributions. The value here is a defensible marker set for non-exhaust source apportionment, which matters more each year as tailpipe PM falls. Caveats are the usual dynamometer ones: WLTP braking is not real-world braking duty, only four couples were tested, and no on-road or roadside validation is attempted.

doi: 10.1016/j.talanta.2026.130448

Fei et al. 2026 Front Immunol

SHAP-interpretable machine learning integrating exposures and multi-omics in COPD-lung cancer comorbidity

A sequential cross-scale framework: GBD epidemiology 1990–2021 across 204 countries, exposome-wide random-forest classification with SHAP interpretation, immune characterisation via CIBERSORT and single-cell RNA sequencing, candidate-gene prioritisation by SMR and LASSO, then wet-lab validation including a macrophage-epithelial Transwell co-culture. **SHAP identified smoking, particulate matter pollution and residential radon as the major shared risk factors**; TREM1 and ODF2L emerged as shared hub genes, downregulated in both diseases, and macrophage-specific silencing of either attenuated macrophage-driven A549 migration and proliferation. The exposure side is GBD-level and coarse — this is not an exposure-assessment paper — but it is a clean worked example of putting an exposome variable through an interpretable-ML pipeline and ending at a testable target.

doi: 10.3389/fimmu.2026.1875146

Filioglou et al. 2026 Atmos Chem Phys*Particle linear depolarisation ratio at 1565 nm from Halo Doppler lidar*

Assesses whether a wind-profiling Halo Photonics StreamLine Doppler lidar can type aerosol at 1565 nm, a wavelength where depolarisation is rarely characterised, across three tropospheric case studies. **Extremely fresh smoke gave 0.017 ± 0.004** , aged long-range-transported smoke marginally higher, and **volcanic ash layers 0.45 ± 0.01** — a separation wide enough to type layers unambiguously. Smoke mass concentration retrieved from the 1565 nm signal agreed with concurrent in-situ observations. The value here is that an instrument already deployed for wind now also delivers aerosol typing and a mass estimate on the same beam; three case studies is thin for a conversion factor, and the aged-smoke result is qualitative.

doi: 10.5194/acp-26-11525-2026

Kalkavouras et al. 2026 Atmos Pollut Res*Ten-year trends in ultrafine particle number concentration and size distribution, Athens*

METADATA ONLY — no abstract was deposited, so no trend, magnitude or instrument detail is asserted here. Flagged because a decade-long urban particle-number and size-distribution record is exactly the reference series that a low-cost optical network cannot reproduce and must be evaluated against; worth retrieving in full.

doi: 10.1016/j.j.apr.2026.103181

Legaria-Santiago et al. 2026 Atmosphere*Leveraging remote traffic data for local air pollutant estimation across London monitoring sites*

Four interpretable tree ensembles (Random Forest, Extra Trees, LightGBM, XGBoost) across six predictor scenarios, from remote traffic + meteorology + temporal variables alone up to including one and four neighbouring stations, benchmarked against ridge regression, spatial interpolation and a cross-site validation. Without neighbouring stations, NO₂ RMSE fell from **9.73–11.66 to 8.72–11.52 $\mu\text{g}/\text{m}^3$** when traffic information was added, and SHAP shows traffic variables contributing at a level comparable to neighbouring-station measurements in traffic-dominated settings. The practical claim — remote traffic counts can partially substitute for network density — is demonstrated for NO₂; the PM channels are reported but not where the gain is.

doi: 10.3390/atmos17080806

Li et al. 2026 Environ Sci Technol*Crowdsourced, community-engaged VOC monitoring and source apportionment near a landfill*

Community citizen-science campaign near the Bristol, Virginia solid-waste landfill using a modified ultrasonic personal air sampler under EPA Method TO-17. BTEX remained below EPA reference concentrations throughout but was significantly elevated relative to comparison locations. Gas-phase, not particulate — carried here for the deployment methodology rather than the analyte. The transferable content is the study-design integration of resident-reported odour events into sampling triggers, which is the same problem a community PM network faces when episodic sources drive the exposure and a fixed duty cycle misses them.

doi: 10.1021/acs.est.6c04657

Ng et al. 2026 Atmos Chem Phys*Impacts of droughts on biomass burning emissions, air quality and public health in the Amazon*

Satellite fire activity plus GFEDv4s emissions driving GEOS-Chem High Performance, 2010–2015, with paired fire-on/fire-off simulations. In non-drought years biomass burning contributes **34.0% (32.0–35.5) of dry-season CO and PM_{2.5}**; in drought years that rises to **55.4% (44.1–76.7)**. Fire-induced PM_{2.5} and ozone raise premature mortality (GEMM) by 6.0% and 18.6% in non-drought years, 8.9% and 24.4% in drought years — *at stable deforestation rates*, which is the point: the climate-driven amplification is separable from the land-use driver. For network design, the drought-year contribution range is wide enough that a fixed seasonal baseline will misestimate the fire fraction by a factor of two.

doi: 10.5194/acp-26-11835-2026

Perišić et al. 2026 Sci Total Environ*Explainable AI resolves ten recurrent environmental settings behind urban PM₁₀ variability*

Four years (2020–2023) of routine regulatory data from a *single* station — PM₁₀, PM_{2.5}, NO₂, SO₂, O₃ and meteorology — fed to ensemble machine-learning models with metaheuristic hyperparameter optimisation, reaching $R^2 = 0.913$, then clustered on SHAP-derived predictor-impact profiles rather than on concentrations. **Ten recurrent settings emerge**. The strongest positive contributions are cold-season accumulation regimes: E0 is frequent and predominantly nocturnal at a mean impact of **50.9 $\mu\text{g m}^{-3}$** ; E7 is rarer but reaches **82.9 $\mu\text{g m}^{-3}$** and persists day and night. Warm, well-mixed settings run **26–29 $\mu\text{g m}^{-3}$ below** the model baseline. The substantive claim — that equal concentrations can encode different pollutant–meteorology configurations — is well supported and directly useful where speciation is unaffordable. The limitation is structural and the authors say so: one station, one climate, no external validation, and SHAP attributions are model-conditional, so the “settings” are properties of a fitted ensemble before they are properties of the atmosphere. For distributed low-cost networks the appeal is that the entire input set is what a modest node plus a met sensor already produces.

doi: 10.1016/j.scitotenv.2026.182195

Rapp et al. 2026 Atmos Meas Tech*Curve fitting algorithm for multimodal particle size distributions: a theoretical basis*

Open-source lognormal decomposition with no predefined mode templates and no mode-count constraint. Each mode's N_i , σ_g and $\bar{D}_{pg}$ are fitted by Levenberg–Marquardt and iteratively subtracted until convergence. Evaluated against a ten-year ambient record at Storm Peak Laboratory and against laboratory PSDs of size-selected ammonium sulfate coated in oxidised biogenic SOA. The result that matters: **a tri-modal assumption is inadequate for 55.7% of new-particle-formation data**, four or more modes being required — so the default configuration of most PSD fitting software is wrong in the majority of the regime it is applied to. Agreement with existing models at a tri-modal setting is strong with substantially lower computation. Supplemental routines format output from common mobility particle size spectrometers, which lowers the adoption cost considerably.

doi: 10.5194/amt-19-5373-2026

Saravagna Sai Lakshmi et al. 2026 J Vis Exp*CORTA-Net: hybrid deep learning with attention-based interpretability for PM_{2.5} forecasting*

CorrXGBoost-Rank feature screening, transfer-learning LSTM encoder, multi-head attention for feature- and timestep-level attribution, over CPCB/DPCC station data, IMD meteorology and MODIS fire counts for Delhi 2012–2024. Reported to reduce prediction error against random forest, XGBoost, LSTM and attention-LSTM baselines — but **no RMSE, MAE or R^2 appears in the abstract for either the model or the baselines**, so the size of the improvement is unstated and unverifiable here. A methods-video venue rather than a modelling journal, and the contribution is an assembly of existing components. Included for the fire-count integration, which is the one input choice that is specific to the Delhi problem rather than generic.

doi: 10.3791/71004

Shlipak et al. 2026 ACS EST Air*Spatial and temporal trends of ambient PM_{2.5} and source-apportioned black carbon in Addis Ababa*

Three continuous years (Apr 2022–Mar 2025) from the MAIA surface network: 10 PurpleAir units across the city calibrated against collocated Beta Attenuation Monitors at two contrasting U.S. Embassy sites (Central, Jacros), plus AethLabs MA350 microaethalometers at those two sites. Network mean PM_{2.5} **30 µg/m³**; BC 3.7 and 7.7 µg/m³; the WHO 24 h guideline of 15 µg/m³ was exceeded on **99% of days**. Diurnal peaks at morning traffic and under nighttime inversion, strongest in the wet season; highest concentrations west of the city centre. Aethalometer-model apportionment attributes **94% of BC to fossil combustion**, 6% to biomass — the opposite of the usual prior for the region. Directional and back-trajectory analysis raises long-range shipping transport from the Gulf of Aden as a regional contributor. Limitation: only two sites carry a reference anchor and BC, so the calibration transfer across the other eight nodes is assumed rather than demonstrated. The single most directly transferable deployment design in this window.

doi: 10.1021/acsestair.6c00140

Wang et al. 2026 Environ Sci Technol*Remote calibration of a nationwide dense low-cost sensor network for hyperlocal air surveillance*

Removes physical collocation from the LCS calibration loop. A spatiotemporal pairing algorithm selects representative training data linking uncollocated sensors to reference stations, with metropolis-specific monthly model refresh; validated on **>33 million hourly observations** across ten US metros. Demonstrated downstream on high-granularity land-use regression and wildfire plume tracking. Two things to hold against it. First, the abstract reports agreement qualitatively and gives *no* R^2 , RMSE or bias, so it cannot be placed against the collocated calibration literature without the full text. Second, and more usefully, the paper quantifies its own denominator problem: LCS density differs significantly by race/ethnicity and income, with African American communities least covered. Remote calibration widens the usable network; it does not move a sensor into a neighbourhood that has none.

doi: 10.1021/acs.est.6c03270

Wei et al. 2026 iScience*Stacked Transformer-XGBoost model with SHAP interpretability for AQI prediction in Linfen*

Transformer temporal encoder stacked with XGBoost through temporally consistent out-of-fold stacking, on 2013–2024 air quality and meteorological series from a heavily industrialised Chinese city. Reported $R^2 = 0.955$, **RMSE 9.05**, **MAE 2.91**; SHAP attributes the variance to PM_{2.5}, NO₂, lagged AQI and seasonal terms. The performance claim is against conventional ML and DL baselines on the same series, with no external site held out, so it is an in-sample-city result. The out-of-fold stacking discipline is the part worth borrowing — naive stacking on autocorrelated air quality series leaks badly.

doi: 10.1016/j.isci.2026.116396

Yang et al. 2026b Atmos Chem Phys*Dust transport versus local emissions in shaping urban ice-nucleating particles*

Month-long winter campaign in Taiyuan, a heavily industrialised Chinese city: immersion-mode INP concentration and ice-nucleation active-site density alongside size distributions and chemical composition. N_{INP} 0.05–13.37 L⁻¹ at and above −15 °C. During a desert-dust event **7.47 L⁻¹ (95% CI 6.64–8.41) versus 0.61 L⁻¹ otherwise**, n_s 1.57×10⁷ versus 9.52×10⁵ m⁻². During pollution periods N_{INP} correlated only weakly with sulfate, nitrate and OC ($|r| < 0.3$), and none of five PMF-resolved PM_{2.5} source factors — industrial, dust, secondary, coal and traffic, fireworks — covaried significantly with it despite dominating the mass. One city, one winter, one dust event carrying the contrast. Still the cleanest recent demonstration that a PM_{2.5} mass record does not index a co-located particle property.

doi: 10.5194/acp-26-11771-2026

Zeng et al. 2026 Atmos Environ*Aerosol vertical distributions determine the seasonal relationships between column AOD and surface PM_{2.5} over Europe*

Metadata only. No abstract deposited. The title states a result that matters directly to satellite-based PM_{2.5} estimation — the AOD-to-surface conversion being governed by vertical structure and therefore seasonally non-stationary — so this is the highest-priority record of the six for abstract retry or full-text retrieval on the next run.

doi: 10.1016/j.atmosenv.2026.122305

Zhang et al. 2026 Environ Sci Technol*Wearable device for direct assessment of particle deposition in the human respiratory system*

Face-mask-format sampler collecting inhaled and exhaled aerosol simultaneously, so deposition is measured by difference rather than inferred from a lung-deposition model. Bench validation confirmed exhaled-particle capture with minimal ambient interference; deployed on **12 healthy volunteers** in subway environments. **~66% of inhaled PM deposited**, highest for PM₁; Fe, Cu and Pb dominated the subway aerosol and deposition varied with lung function and exercise habit. Twelve subjects in one microenvironment is a feasibility study, not a dosimetry dataset, and the size-resolved difference method is vulnerable to hygroscopic growth of exhaled particles at 100% RH — which the abstract does not address. The device concept is nonetheless the most directly relevant instrumentation record of the issue: it measures the delivered dose that every ambient network is a proxy for.

doi: 10.1021/acs.est.6c05543

Zhao et al. 2026 J Aerosol Sci*Monte-Carlo inversion of aerosol complex refractive index from multi-wavelength lidar*

Metadata only. Elsevier returned no abstract through Crossref and ScienceDirect did not serve the record; no finding is asserted. The registered title describes an improved Monte-Carlo sampling algorithm that inverts the aerosol complex refractive index from multi-wavelength lidar signals combined with an independently measured particle size distribution. Carried at tier C because refractive index is the parameter that converts an optical measurement into a mass estimate, and every low-cost nephelometric sensor assumes one implicitly; whether this inversion is well-posed cannot be judged from the title.

doi: 10.1016/j.jaerosci.2026.106883

Zundel et al. 2026 (preprint) medRxiv*Wearable personal PM_{2.5} monitoring during wildfire smoke episodes in Metro Detroit*

AirBeam3 wearables on four participants (one adolescent, three adults) during southeast Michigan smoke episodes, with neighbourhood outdoor PM_{2.5} taken as the mean of the three nearest PurpleAir units and smoke days identified from state advisories. **Personal 28.2±20.0 vs 5.1±4.7 µg/m³** on smoke vs non-smoke days; neighbourhood **87.6±80.2 vs 12.3±7.0**. Within participants, each 10 µg/m³ of neighbourhood PM_{2.5} mapped to only **2.3 µg/m³** personal — personal exposure is both attenuated and poorly tracked by the outdoor proxy, yet still put >25% of participant-days over the EPA 24 h standard. $n = 4$ and no formal uncertainty on the slope: this is a feasibility demonstration, and the attenuation factor should not be reused as a calibration constant. Preprint, not peer reviewed.

doi: 10.64898/2026.07.30.26359253

Exposure assessment & modelling

36 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (36 OF 36 RECORDS IN THIS SUBTOPIC)

Abadi et al. 2026 Atmos Res*Future projections of surface dust concentrations in major dust source regions of the Mesopotamian plains***Metadata only.** Crossref sweep, no abstract deposited. Projected mineral-dust concentrations for a source region that exports coarse PM across the Gulf and into South Asia; no scenario, horizon or model family is extractable from the deposited metadata.

doi: 10.1016/j.atmosres.2026.109282

Alaimo et al. 2026 Atmosphere*Elemental composition and Pb isotopic signatures in pine needles from the industrialized Milazzo area*Forty *Pinus pinea* needle samples by ICP-OES/ICP-MS across 36 elements, 25 of them also for $^{206}\text{Pb}/^{207}\text{Pb}$ and $^{208}\text{Pb}/^{206}\text{Pb}$. Enrichment factors above 10 for Cd, Cu, Mo, Sb, V and Zn against EF 0.5–2 for Al, Fe and Ti clearly separate anthropogenic from geogenic contributions. Pb ratios of 1.153–1.192 and 2.063–2.108 indicate mixing of industrial emissions with local geological background and, notably, *limited* residual leaded-gasoline signal. A good template for isotope-assisted source apportionment where no monitoring network exists.

doi: 10.3390/atmos17080805

Albugami 2026 Atmosphere*Dust event characteristics, transport pathways and source regions over Riyadh*34 dust events over 2024–2025 identified from AERONET by a dual optical criterion ($\text{AOD}_{500} \geq 0.50$ and 440–870 nm Ångström exponent ≤ 0.50), cross-checked against NOAA ISD present-weather and visibility and traced with HYSPLIT back-trajectories, PSCF and CWT. **82% fell in March–May**; mean Ångström exponent 0.28, mean fine-mode fraction 0.26. Surface present-weather codes confirmed 26 of 34; the other 8 (24%) were classified as elevated dust invisible at the surface station — a clean demonstration that column and surface instruments disagree by design, not by error. Source–receptor analysis points to the Mesopotamian Basin, with the northern Rub’ al-Khali secondary. Single-author, single-station, two years.

doi: 10.3390/atmos17080802

Alshehri & El-Hamid 2026 (preprint) Research Square*Sentinel-5P and MODIS land-surface coupling of pollutant fields over Riyadh, 2018–2025*Eight years of TROPOMI NO_2 , SO_2 , CO, O_3 and Absorbing Aerosol Index combined with MODIS land-surface temperature and NDVI and Sentinel-2 land cover in Google Earth Engine, analysed by correlation, PCA and multiple linear regression. NO_2 and CO concentrate over urban and industrial areas, O_3 peaks downwind, LST falls from 43.10 °C (2021) to 37.07 °C (2025) while NDVI rises after 2022. **The regression is weak: R^2 ranges from 0.124 (O_3) to 0.231 (SO_2)**, and PCA explains 55.8% of variance; the authors correctly conclude that land-surface descriptors explain only a minor share and that emissions and transport dominate. Carried as a negative methodological result for an arid-region setting where ground networks are sparse — and as a caution, because **no surface PM product appears anywhere in the study** despite the framing as air pollution monitoring; AAI is a column absorption index, not a PM concentration. Preprint, not peer reviewed, and the LST trend is reported without any test for its significance against interannual variability.

doi: 10.21203/rs.3.rs-10538140/v1

Ashta et al. 2026 Atmos Chem Phys*Wet and dry atmospheric deposition of microplastics across a Swiss site gradient*Four-weekly wet and dry deposition sampling over one year (May 2024–May 2025) at urban (Zurich), suburban (Dübendorf), rural (Magadino, Payerne) and mountainous (Chaumont) sites, with focal-plane-array $\mu\text{-FTIR}$ over the 20–215 μm range and an explicit measurement-uncertainty treatment. Urban deposition **881 $\text{MPs m}^{-2} \text{d}^{-1}$ (95% CI 562–1199)** and **53 $\mu\text{g m}^{-2} \text{d}^{-1}$ (17–107)**; the other four sites are similar to each other at 249–331 $\text{MPs m}^{-2} \text{d}^{-1}$ (140–478). Scaled to land use, 219 t or 3.8×10^{14} particles per year below 2000 m nationally, excluding tyre wear. The 20 μm lower cut is the binding limitation: the inhalable fraction sits mostly below it.

doi: 10.5194/acp-26-11803-2026

Awan et al. 2026 Aerosol Air Qual Res*BRIQ- $\text{PM}_{2.5}$: quantile-mapping attribution of industrial excess $\text{PM}_{2.5}$ across Pakistan*Proposes estimating industrial impact as the residual between an observed satellite $\text{PM}_{2.5}$ distribution and the same distribution quantile-mapped onto a province-specific "low-industry" baseline city, then interpolating the residual by variogram modelling and universal kriging across four provinces. Reported validation is distributional — histogram and ECDF alignment, improved RMSE after correction. **The attribution is definitional, not estimated**: whatever separates a target city from its chosen baseline is labelled industrial, so meteorology, dust, biomass burning and population density all load onto the same term, and no ground-monitor comparison or sensitivity analysis over baseline choice is reported. Useful as a hotspot screen; not an emissions attribution.

doi: 10.1007/s44408-026-00160-z

Blanco-Alegre et al. 2026 Atmos Chem Phys*Emission factors and particle morphology from prescribed fires in Mediterranean shrubland*Field campaign on *Calluna vulgaris* and *Genista hispanica* burns in north-west Spain combining gravimetry, thermal–optical transmittance, ion chromatography, FTIR and SEM. *Calluna* burned flaming (MCE 90.6%), *Genista* smouldering (MCE 70.8%); gas-phase concentrations were higher for *Calluna* but emission factors for incomplete-combustion products were higher for *Genista*. **$\text{PM}_{2.5}$ EFs 53.5 g kg^{-1} (*Calluna*) and 44.7 g kg^{-1} (*Genista*)**, with OC and EC at 28.1% and 32.9% of $\text{PM}_{2.5}$ mass and water-soluble inorganic ions at 6.9% and 4.5%. SEM showed abundant tar-ball-like submicron particles and thickly coated aggregates. Two vegetation types at one site — the EFs are not a regional average — but combustion-regime-resolved and directly usable.

doi: 10.5194/acp-26-11695-2026

Cai et al. 2026 Atmos Environ*Effect of ammonia on reactive oxygen species generation in secondary organic aerosols***Metadata only.** Crossref sweep, no abstract deposited, and no study location stated. Ammonia-modulated oxidative potential of secondary organic aerosol sits precisely between the previous issue’s Nanjing ammonia-thermodynamics result and today’s oxidative-priming mechanism papers, so it is the second-highest-priority retry after Zeng et al.

doi: 10.1016/j.atmosenv.2026.122307

Chai et al. 2026 Environ Pollut

Role of ammonia, aerosol pH and liquid water content in secondary inorganic aerosol formation in suburban Nanjing

Hourly water-soluble ions and precursor gases, January–March 2021, coupled to ISORROPIA-II and WRF-Chem. SIA is **49.55% of PM_{2.5} mass** (mean 34.4 $\mu\text{g}/\text{m}^3$), nitrate-dominated. Continuous ammonia excess fully neutralises the anion pool; rising aerosol liquid water enhances aqueous oxidation and drives nitrate and ammonium into the particle phase, while pH governs both semi-volatile partitioning and the sulfate pathway (transition-metal-ion catalysis dominant). Thermodynamically the system is **total-nitrate-limited**: PM_{2.5} responds minimally to moderate NH₃ cuts and needs >60% reduction to move, whereas cutting HNO₃ alone matches joint HNO₃+NH₃ control. The policy inversion — prioritise NO_x, treat NH₃ as a local adjunct — is the transferable result.

doi: 10.1016/j.envpol.2026.128998

Chuang et al. 2026 Atmos Res

Simulating PM_{2.5} in southern Taiwan under high-pressure return flow (Kao-Ping Experiment)

METADATA ONLY — a KPEX modelling study of a synoptic regime that traps PM_{2.5} over the Kao-Ping region. No abstract in the deposit and no indexed full text on 20 August. No quantity inferred.

doi: 10.1016/j.atmosres.2026.109286

Defrance 2026 (preprint) Research Square

Beyond the fire perimeter: the smoke footprint that constrains outdoor activity dwarfs the burned area of the July 2026 Gironde–Landes wildfire episode

ERA5-Land wet-bulb globe temperature combined with CAMS analysis PM_{2.5} at hourly resolution over 25–31 July 2026, against 40 background monitoring stations. The two fires burned 340 km² and forced ~220,000 evacuations; the footprint above the European AQI “Poor” threshold (hourly PM_{2.5} > 50 $\mu\text{g}/\text{m}^3$, where reduced outdoor activity is advised) reached **149 times the burned area** and about **370 km downwind**, with 139× on the EU 2030 daily limit of 25 $\mu\text{g}/\text{m}^3$ and at least 121× against the larger operational fire perimeter. Plume co-location rose from 73% hourly to 94% daily. At the stations, a same-day heat–smoke relay affected 28% of affected station-days at 25 $\mu\text{g}/\text{m}^3$ and 19% at 50, against 0–2% in the co-located grid cells — **the 10 km analysis smooths away the peaks that drive the relay**, and reproduces the episode at daily $r = 0.83$ while underestimating peaks by a median factor of ~2.2. Preprint, single author, single episode; the self-directed error analysis is unusually thorough for the format.

doi: 10.21203/rs.3.rs-10757961/v1

Dias et al. 2026 Bull Environ Contam Toxicol

Micro-scale spatial patterns of PM_{2.5} across urban green and non-green areas in an arid city

Six mobile monitoring campaigns along a fixed 4.3 km transect with **75 georeferenced sampling points** in San Juan, Argentina, winter 2026, classified green versus non-green by surrounding vegetation cover. Generalised linear mixed models adjusting for traffic intensity, distance to nearest road and repeated measurement give **~31% lower adjusted PM_{2.5} in green spaces**. The adjustment set is the weak point — traffic intensity and distance-to-road are the obvious confounders but green spaces in an arid city also differ in surface dust availability and irrigation, which is an unadjusted pathway that would produce the same sign. A single winter and a single transect. Useful mainly as a demonstration that mobile monitoring resolves within-block gradients that a regulatory network cannot, in a South American arid setting where almost no such data exist.

doi: 10.1007/s00128-026-04318-8

Do et al. 2026 Environ Sci Technol

Field-constrained diurnal viscosity of urban PM_{2.5} and implications for N₂O₅ uptake

Poke-and-flow measurement with fluid-dynamics inversion on PM_{2.5} collected in Ansan, South Korea (summer 2024): **6.2×10⁴–1.3×10⁷ Pa s at 20–40% RH**, exceeding 10⁸ Pa s below ~10% RH. Pooled with Seoul and Beijing data into a Vogel–Tammann–Fulcher temperature–RH parametrisation and pushed through an ML surrogate over hourly meteorology for 14 megacities. Predicted viscosity is systematically lower at night everywhere (RH plasticisation beating nocturnal cooling), giving nighttime N₂O₅ uptake coefficients of **0.01–0.07** against the conventional liquid assumption of 0.1. Weakness stated by the authors: the parametrisation rests on 24 h composition, so no time-resolved chemistry enters the diurnal prediction it is being used to make.

doi: 10.1021/acs.est.6c04811

Dovrou et al. 2026 Environ Sci Atmos

Wintertime wildfires reshape urban atmospheric chemistry: the January 2025 Los Angeles fires

METADATA ONLY — the deposited abstract is truncated before any result. Verifiable from it: the study concerns how dominant chemical regimes and meteorology govern wildfire air-quality impacts in urban coastal regions, using the January 2025 Los Angeles fires. Flagged for retrieval given how much of this watch’s recent health signal has come from source-resolved wildfire PM_{2.5}.

doi: 10.1039/d6ea00096g

Duan et al. 2026 Atmos Chem Phys

Regime-dependent aerosol pH in ammonia-rich urban Beijing from explainable machine learning

Field observations plus thermodynamic modelling, interrogated with interpretable ML to apportion pH variability. Temperature dominates, at roughly **–0.6 pH units per 10 °C**. Excess ammonia, nitrate-to-sulfate ratio and PM₁ mass are positively associated with pH, with strongest sensitivity at low values and saturation at high ones. Relative humidity is the interesting one: its sign *reverses* by regime — higher RH increases acidity below ~15 °C, at N/S > ~1.25, or at PM₁ > ~50 $\mu\text{g}/\text{m}^3$, and decreases it under warm, sulfate-dominant or low-mass conditions. Feature attribution is association, not causation, and pH is itself a model output rather than a measurement. The RH regime reversal is a useful caution for anyone applying a single humidity correction across seasons.

doi: 10.5194/acp-26-11683-2026

Fan et al. 2026 iScience

Multiple amides synergistically enhance sulfuric acid–ammonia nucleation

Quantum-chemical cluster thermodynamics plus kinetic simulation of new particle formation in binary, ternary and quaternary systems containing sulfuric acid (SA), ammonia, methylacetamide (MAA) and methylpropionamide (MPA). **(SA)₁·(MPA)₁ and (SA)₁·(MAA)₁ are more stable than (SA)₁·(NH₃)₁ and even than (SA)₁·(DMA)₁**, and the two amides together enhance nucleation by **an order of magnitude more than either alone**, most strongly at low SA concentration and low temperature. A displacement hierarchy is identified: MPA replaces NH₃, MAA replaces MPA. Carried because secondary particle number — not mass — is what an optical low-cost sensor is blindest to, and because amide sources (industrial, agricultural, carbon-capture solvent degradation) are rising and largely unmonitored. It is a purely computational result with no chamber or ambient confirmation, the amide concentrations assumed are not tied to measured ambient values, and cluster-level thermodynamics has a poor track record of quantitative transfer to observed formation rates.

doi: 10.1016/j.isci.2026.117013

Gonçalves et al. 2026 Atmos Pollut Res*Extreme Saharan dust outbreak: air quality crisis in NW Spain (March 2022)*

Metadata only. Surfaced through the Crossref journal sweep with no abstract deposited. Title, journal, DOI and authorship are verified; design, sample size and quantitative outcome are not extracted and are not inferred. Flagged for abstract retry on the next run.

doi: 10.1016/j.apr.2026.103183

Guo et al. 2026 Atmos Environ*Meteorology-driven long-range transport to an ultra-low emission region during the monsoon*

Metadata only. Elsevier withheld the abstract; Semantic Scholar returns 404 for the DOI. The registered title indicates a Hainan case study of monsoon-driven long-range transport into a low-background region. Nothing beyond the title is asserted. Transport-into-clean-background studies are the useful test case for source attribution because the local emission term is small, which is why this is carried rather than rejected — but it carries no extractable content until the abstract is released.

doi: 10.1016/j.atmosenv.2026.122293

Ianiri et al. 2026 Environ Pollut*Street- versus roof-level deposition fluxes of POPs and metals, normalised to sedimentable PM*

Thirteen monthly deposimeter samples at two heights on one Rome building, Aug 2023 – Aug 2024, with PCDD/Fs, PCBs and PAHs by GC-HRMS and metals by ICP-MS, reported both as fluxes and normalised to collected sedimentable PM. **Annual SPM flux 42 vs 26 mg m⁻² d⁻¹** street versus roof, but a Wilcoxon signed-rank test found **no significant street–roof difference in SPM-normalised concentrations** ($p > 0.05$): height changes deposited quantity, not deposited composition. Seasonality was significant for PCDD/Fs, PCBs, Cr, As, Ni and Sb (winter peaks) and for PAHs (summer/autumn). Single building, single city, $n = 13$ per height — a design that supports the composition claim far better than the flux ratio.

doi: 10.1016/j.envpol.2026.128965

Jin et al. 2026 Atmos Pollut Res*Hourly 2 km Himawari-8 aerosol fine-mode fraction retrieval improves PM_{2.5} mapping*

Metadata only. Elsevier withheld the abstract and the record is too recent for Semantic Scholar (404) or Scholar Gateway, whose corpus currently ends in May 2026. The registered title asserts that adding a fine-mode-fraction retrieval to Himawari-8 geostationary observations improves surface PM_{2.5} estimation over the Beijing–Tianjin–Hebei region at hourly, 2 km resolution. **No performance figure, validation design or comparison baseline is asserted here, because none was available.** Flagged for retrieval on a later run. Geostationary fine-mode retrieval is the most direct satellite analogue of the composition-resolution problem raised elsewhere in this issue, so the numbers are worth chasing.

doi: 10.1016/j.apr.2026.103180

Joshi et al. 2026 (preprint) Research Square*Environmental drivers of foliar particulate accumulation, leaf structure and biochemistry in urban trees in Delhi*

Ten native species, four contrasting Delhi sites, summer and winter 2025, with branch-level triplicates. Total leaf-deposited PM rose from **120.13 ± 30.99 to 423.69 ± 30.99 μg cm⁻²** between seasons, fine PM from 6.82 ± 1.32 to 25.75 ± 1.32 . *Plumeria alba* retained most (414.92), *Azadirachta indica* least (140.82). Higher loads tracked ascorbic acid and foliar pH positively and carotenoids negatively, and were *not* explained by specific leaf area, leaf dry matter content or thickness; APTI showed no seasonal change. The useful negative result is that capture capacity and physiological tolerance are separate axes, so single-metric species screening is inadequate.

doi: 10.21203/rs.3.rs-10524655/v1

Kan et al. 2026 Atmosphere*Short-term observations of airborne microplastics in Phnom Penh: concentrations, aerodynamic size distribution and polymer composition*

First AMP observations for Cambodia, by μFTIR-ATR imaging, with meteorology, gaseous co-pollutants, water-soluble ionic tracers and HYSPLIT back-trajectories. Four independent 72-h periods: **0.97 ± 0.30 MP m⁻³ in TSP** (range 0.55–1.27) and **0.33 ± 0.10 MP m⁻³ in PM_{2.5}** (0.23–0.49); 134 particles identified in TSP, 46 of them inside the PM_{2.5} fraction. PE, PP and PET dominate, 52% fragments, 82% below 30 μm Feret diameter. The authors are explicit that $n = 4$ precludes source attribution — read the polymer split as a baseline, not an apportionment.

doi: 10.3390/atmos17080804

Kommula et al. 2026 Environ Sci Technol*Comparable organic aerosol mass enhancement during photochemical and dark oxidation of biomass burning emissions*

Chamber aging of emissions from savannah grass, savannah wood and boreal forest surface fuel under varied combustion conditions, comparing daytime photochemical with nighttime dark oxidation. OA is 82–99% of PM₁ in the fresh emissions. Both aging routes yield **comparable net OA mass**, which PMF traces to more efficient primary-OA loss during daytime offsetting more efficient gas-phase oxidation. The resulting mass enhancement of **0.75–1.3×** matches field observations and closes a long-standing lab–field discrepancy. Applies to low-NO_x remote conditions; the authors say so, and that is where the proposed model simplification is and is not safe.

doi: 10.1021/acs.est.5c15841

Lazo et al. 2026 J Environ Qual*Biomonitoring trace metal deposition with Ligustrum lucidum: Albanian fire incidents*

Passive biomonitoring across 14 urban, industrial and rural sites in Elbasan, leaves collected two weeks after an industrial-waste fire and a landfill fire. Al, V, Fe, Cu, Cd, Pb and Zn showed low-to-moderate variability; Cr, Co, Ni, As and Sb were highly variable with industrial-site outliers, and cluster analysis separated the two groups. Method demonstration rather than exposure assessment — there is no control period and no airborne concentration — but it is the kind of post-incident evidence that is otherwise unobtainable.

doi: 10.1002/jeq2.70243

Liu et al. 2026 Atmos Res*Identification of dominant weather patterns and circulation regulation mechanisms of winter PM_{2.5} pollution in Iran*

Metadata only. Crossref sweep, no abstract deposited. Synoptic classification of winter PM_{2.5} episodes; one of very few Iranian PM records the watch has surfaced, which is why it ships as a stub rather than being dropped.

doi: 10.1016/j.atmosres.2026.109287

Mahia et al. 2026 J Environ Manage

Street geometry, air quality and real-time physiological response in elderly pedestrians

Fourteen elderly participants walked five routes in Perugia during spring while wearable sensors logged physiology and portable stations logged microclimate and air quality. Paths with higher height-to-width ratios (0.92–1.27) and elevated NO₂ and PM₁₀ (to 21 ppb and 18.77 µg/m³) significantly degraded air-quality perception ($\beta = -0.273$, $p = 0.028$ for NO₂; $\beta = -0.312$, $p = 0.013$ for PM₁₀), and both pollutants tracked skin-conductance change and self-reported emotional state; higher Green Visual Index went with more relaxed responses. With $n = 14$ over one season the coefficients carry little weight, and perception and physiology are both plausibly driven by street geometry directly rather than by the pollutant. Read it for the measurement stack — paired personal air-quality and physiological logging at walking cadence — not the estimates.

doi: 10.1016/j.jenvman.2026.130681

Moryani et al. 2026 J Appl Toxicol

PM_{2.5}-bound Cu, Cd, Ni, Pb and Zn across 20 schools in two contrasting South China cities

Paired summer and winter sampling at 20 schools in Guangzhou and Maoming, ICP-MS for five metals, with geoaccumulation index, enrichment factor and PCA source apportionment, then inhalation risk assessment for children and adults. Cu and Cd carried the highest enrichment (EF 27.72 and 27.32 in Guangzhou), winter exceeded summer, and PCA pointed to traffic, industrial combustion and urban emissions. **All hazard quotients and carcinogenic risks fell below accepted thresholds**, with Pb (4.85×10^{-3}) and Cu (3.27×10^{-3}) leading non-carcinogenic risk and Cd (7.30×10^{-6}) carcinogenic. Fieldwork dates from 2018, so it characterises an emissions regime seven years old; the null risk result is the honest headline and is under-stated in the abstract.

doi: 10.1002/jat.70398

Naghizade et al. 2026 Atmos Environ

Multi-class organic marker profiling of urban PM₁₀ in Strasbourg

METADATA ONLY — seasonal distribution of exhaust, non-exhaust and biomass-burning tracers in northeastern France. No abstract in the Crossref deposit; Semantic Scholar and Europe PMC both returned nothing on 20 August. No quantity is inferred here. Flagged for the next issue that can reach the full record.

doi: 10.1016/j.atmosenv.2026.122306

Qing et al. 2026 Atmos Pollut Res

Seasonal and urban-suburban characteristics, emission and spatial sources of PM_{2.5}-bound PAHs and n-alkanes in Wuhu City

Metadata only. Crossref sweep, no abstract deposited. A source-apportionment study of the particle-bound organic fraction, which is the composition class the previous issue's mortality result pointed at — flagged for retry rather than characterised here.

doi: 10.1016/j.apr.2026.103186

Saarikoski et al. 2026 Atmos Chem Phys

Role of hydrocarbons and oxygenated organic species in traffic-derived aerosol

Four Aerodyne AMS campaigns (2018–2024) at a Helsinki traffic site with PMF source apportionment. Beyond hydrocarbon-like OA peaking at the morning rush, a second traffic-related factor (TrOA) peaks in the morning but stays elevated longer, with mass spectra between HOA and biomass-burning OA and distinct C₂H₄O₂⁺ (m/z 60), C₂H₅O₂⁺ (61) and C₃H₅O₂⁺ (73) fractions. **Using HOA alone to estimate traffic primary OA underestimates it by ~50%.** The authors are explicit that TrOA's origin — atmospheric processing versus direct emission from modern vehicles — is unresolved, which is exactly the ambiguity that makes the 50% figure a bound rather than a correction factor. Directly relevant to any speciated or source-resolved sensing product.

doi: 10.5194/acp-26-11817-2026

Wang Q et al. 2026 Atmos Chem Phys

Quantitative analysis of aerosol acidity and its driving factors in the Guanzhong area

One year of water-soluble inorganic ions in PM_{2.5} on the semi-arid Guanzhong Plain, with thermodynamic pH reconstruction and sensitivity-based driver attribution. Annual mean pH 3.8 ± 1.0 (winter > spring > summer > autumn), shifting from 2–5 on clean days toward 3–6 on polluted days. Temperature contributes 22.3–33.8% of pH variation across all seasons, NH_x 11.4–44.9% (dominant in autumn/winter), sulfate 8.5–10.8%; Ca²⁺ is a spring-specific driver. RH shows a non-monotonic effect with an inflection at 60–85% RH, attributed to deliquescence-driven liquid water content. Attribution rests on a thermodynamic model rather than measured pH, which is the standing caveat for this whole literature.

doi: 10.5194/acp-26-11785-2026

Wang T et al. 2026 Environ Sci Technol

Chlorine radical cycling on photoactive particulate matter amplifies photolytic NO_x release

Bench photochemistry showing chloride on photoactive PM surfaces oxidised by photogenerated holes to Cl•, which accelerates conversion of nitrite intermediates to gaseous NO_x while chloride is regenerated. **~40-fold NO_x enhancement** under the model conditions, robust across humidity, chloride salt, substrate (including soot) and nitrogen source (ammonium, 4-nitrophenol). Rate scales with photoactive-matter loading and PM composition. Extrapolated to **0.7–6.2% of NO₂ variability** in the study region — a small number derived from a large laboratory enhancement, and the gap between the two is the result's main uncertainty.

doi: 10.1021/acs.est.6c06768

Yang et al. 2026 Atmos Chem Phys

Impacts of anthropogenic aerosols on a snowfall event in the Guanzhong Basin and surrounding areas

Cloud-resolving fully coupled WRF-Chem, with emission-perturbation experiments separating aerosol–radiation from aerosol–cloud interactions. The precipitation response reverses sign between the basin and its surroundings: inside the basin, excluding ARI gives a slight precipitation increase with emissions via enhanced ice-phase processes, and ARI itself increases precipitation once the emission increase passes a threshold (higher relative humidity, larger ice water path, better survival of falling ice); outside, precipitation falls with emissions as ACI suppresses the liquid phase. Spatial distribution is largely insensitive. Relevant to anyone using precipitation as a wet-deposition assumption in a polluted basin.

doi: 10.5194/acp-26-11923-2026

Yu et al. 2026 Environ Sci Technol

Probabilistic assessment of PM_{2.5} and ozone co-occurring extremes under wildfire smoke

Spatiotemporal Bayesian neural field (ST-BayesNF) integrating satellite smoke-coverage products, ground PM_{2.5} and ozone, and meteorological reanalysis to give daily probabilities of *joint* extremes at 5 km over the contiguous US, 2004–2023. The model discriminates co-extreme from non-co-extreme days and is most sensitive to smoke density interacting with boundary-layer height and near-surface temperature; predicted co-occurrence risk rises during smoke events not only in the western US but increasingly in the Midwest and Northeast under long-range transport. No health outcome is modelled — the health premise (co-occurrence raises risk beyond either pollutant alone) is imported from prior literature and not tested here.

doi: 10.1021/acs.est.6c04403

Zaman & Mim 2026 Environ Sci Atmos*Decadal variation of atmospheric PM_{2.5} over Bangladesh from MERRA-2 reanalysis*

A 25-year reanalysis climatology of PM_{2.5} and its speciated components across contrasting Bangladeshi microenvironments, presented by the authors as the first comprehensive treatment for a country that ranks among the most PM_{2.5}-polluted globally. The Crossref abstract is truncated at source after the framing sentence, so trend magnitudes, speciation shares and the validation against ground observations are all unavailable here. The methodological caveat is generic but load-bearing for this kind of study: MERRA-2 assimilates AOD, not surface PM, and its skill over South Asia is known to be poorer than over North America or Europe — so a reanalysis-only trend for Bangladesh needs its ground-truth comparison read before the numbers are used.

doi: 10.1039/d6ea00108d

Zheng et al. 2026 Atmos Res*Aerosol modulation of precipitation systems migrating from mountains to plains in Northern China during early autumn*

Metadata only. Same provenance and same caveat as the entry above. The topic is adjacent to Yang et al. on this page, but nothing in the two can be compared until an abstract or full text is available.

doi: 10.1016/j.atmosres.2026.109289

Cardiovascular & metabolic**3 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (3 OF 3 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026 Ecotoxical Environ Saf*Pollutant-specific plasma proteomic signatures and atrial-fibrillation-free life lost*

UK Biobank, ~2,900 Olink plasma proteins screened against long-term PM_{2.5}, PM₁₀, NO₂, NO_x, SO₂, benzene and O₃ under Bonferroni correction, then elastic-net regularisation to build one signature per pollutant; incident AF over a median **13.53 y** by Cox regression. Signatures out-predicted the external exposure metrics and stratified risk after adjusting for ambient concentration; high-risk participants lost an estimated **1.15–1.25 AF-free years**. High-throughput mediation nominates **ANGPT2, GDF15 and PLAUR**. The weakness is structural, not analytic: a signature trained on a modelled exposure inherits that model's error and its spatial confounding, so "out-performs the exposure metric" may mean "encodes residence better". No HR or interval is reported, so nothing enters today's forest plot.

doi: 10.1016/j.ecoenv.2026.120666

Liu et al. 2026b Indoor Air*Household clean cooking fuel transition and attenuated cardiopulmonary function decline in middle-aged and older adults*

CHARLS 2011–2020, with two independent subcohorts built from baseline solid-fuel users: a short-term physiological cohort (n = 6,534; 27.4% transitioned by 2013) and a long-term disease cohort (n = 6,362; 33.5% transitioned by 2015). Transition attenuated peak-expiratory-flow decline by **11.0 L/min** (95% CI 4.7–17.3) and lowered systolic blood pressure modestly ($\beta = -1.7$, $p = 0.017$). Long term: **OR 0.80 (0.7–1.0, $p = 0.044$) for chronic lung disease** and **0.80 (0.7–1.0, $p = 0.028$) for heart disease**, with no association for hypertension; protective effects were larger in non-smokers and non-drinkers. The two-subcohort design is what lets the physiological intermediate and the disease endpoint be read together.

doi: 10.1155/ina/8898458

Nong et al. 2026 PLoS One*Long-term air pollution and microvascular complications in diabetes*

UK Biobank, 9,671 participants with diabetes, 12.4-year median follow-up, 2,104 (21.8%) incident microvascular complications. Cox models adjusted for demographic, clinical and biochemical covariates give PM₁₀ **HR 1.44 (1.27–1.64)** and NO₂ **HR 1.27 (1.12–1.45)** for the highest versus lowest exposure quartile, stable across stratified and sensitivity analyses, with early divergence of the cumulative-incidence curves. Two caveats: PM_{2.5} is listed among the exposures but no PM_{2.5} estimate is reported in the abstract, and UK Biobank exposure contrasts are narrow, so the quartile framing hides how small the underlying concentration difference is. Restricted cubic splines are reported as showing risk beginning at relatively low levels — the claim that matters for guideline-adjacent argument, and the one the abstract quantifies least.

doi: 10.1371/journal.pone.0354711

Neuro / mental health**4 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (4 OF 4 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026 Environ Toxicol Pharmacol*Integrated plasma proteomics and metabolomes reveal potential biomolecules linking PM_{2.5} to depression*

24,940 depression-free UK Biobank participants, exposure to PM_{2.5} and five constituents (ammonium, nitrate, sulfate, elemental carbon, organic carbon), Olink proteomics and NMR metabolomics, Cox models with mediation, functional enrichment and drug-repurposing prediction. **982 incident cases over a mean 13 years**; HRs reported "up to 1.20 per SD" with ammonium the strongest constituent. **140 proteins and 3 metabolites** were statistically significant mediators, mediation proportions up to 8.00% and 3.42%, enriched in lipid and unsaturated fatty acid metabolism, immune regulation and inflammation. Note that the abstract gives no interval for the 1.20 figure, which is why it does not appear on this issue's forest plot.

doi: 10.1016/j.etap.2026.105144

He et al. 2026 Ecotoxical Environ Saf*Plasma proteomic signatures modifying the air pollution–epilepsy association*

Proteome-wide interaction screen: 51,428 participants, 2,254 Olink-style plasma proteins, PM_{2.5}/PM_{coarse}/NO₂/NO from land-use regression, plus a PCA-derived composite air-pollution index. Logistic models for prevalent and Cox models for incident epilepsy identify **CIT-PM_{2.5}, PTPRR-NO₂ and BCL2L15-API interactions**, enriched for immune, inflammatory and cell-death pathways, with hippocampal-volume and white-matter-hyperintensity correlates and neuronal/endothelial single-nucleus enrichment. The threat to inference is multiplicity: ~2,254 proteins × four exposures with no corrected significance level or interaction magnitude reported in the abstract. A candidate list, not a mechanism. Relevance here is that it is exposure *misclassification* in the LUR layer, not protein measurement, that would most easily generate this pattern.

doi: 10.1016/j.ecoenv.2026.120669

Parmar et al. 2026 Environ Int*Short-term PM_{2.5} and its chemical components and neuropsychiatric hospitalisations*

Ground observations fused with 5 km WRF-Chem output to build a gridded PM_{2.5} field for Delhi 2016–2019, then GAM with distributed lags against hospital neuropsychiatric admissions. Total mass gives **+0.926% (95% CI 0.097–1.762) per 10 µg/m³; black carbon gives +11.509% (3.272–20.402)**, ahead of organic carbon and other inorganics, and only industrial-source PM_{2.5} was significant among source contributions. Among disorder categories only anxiety reached significance. The per-unit-mass potency gap between BC and total mass is the headline, but it should be read with the exposure ranges in mind: BC concentrations are roughly an order of magnitude smaller than total mass in Delhi, so a per-10 µg/m³ contrast for BC is a large extrapolation beyond the observed spread. Single-hospital case ascertainment also limits generalisation across a city of Delhi's size.

doi: 10.1016/j.envint.2026.110464

Perera et al. 2026 Environ Health*Declining prenatal air pollution and improving cognitive scores across successive NYC cohorts*

Three successive Columbia CCCEH birth cohorts, 1998–2020, **n = 1,109**, with prenatal PM_{2.5}/NO₂ (first two cohorts) and airborne PAH (all three), and Bayley MDI at ages 1, 2 and 3. Exposures fell significantly across cohorts and MDI rose significantly; pooled associations were inverse and significant at all three ages, and cohort-specific inverse associations were clearest in the earliest, highest-exposure cohort (*n* > 700) and weaker in the 2008–2016 cohort. The design is the interesting part — using secular exposure decline as the contrast — and also the vulnerability: everything else about being a child in New York changed between 1998 and 2020, and period effects on Bayley scores are not identifiable from cohort effects here. The authors frame it as policy-relevant evidence; it is better read as consistency between two trends than as an estimated effect.

doi: 10.1186/s12940-026-01328-1

Respiratory & allergic**9 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (9 OF 9 RECORDS IN THIS SUBTOPIC)

Ali et al. 2026 Public Health Chall*Symptom burden among adult outpatients during Lahore's 2024–2025 winter smog*

400 outpatients at a tertiary hospital, October 2024 – January 2025, structured questionnaire plus city-level air quality (mean AQI 180, mean PM_{2.5} 96 µg/m³; PM_{2.5} and PM₁₀ above the WHO 24 h guideline on every monitored day). 98% reported at least one symptom — breathlessness 74%, eye irritation 56%, cough 53%. Regression on a composite vulnerability index: **smoking IRR 1.15 (1.00–1.32), routine mask use IRR 0.96 (0.84–1.09), i.e. null**. Exposure is assigned at city level to a health-seeking sample with no unexposed comparison, so no dose–response can be read from this; its value is documenting symptom prevalence in a South Asian megacity where hospital-based evidence is thin.

doi: 10.1002/puh2.70349

Erickson-DiRenzo et al. 2026 J Voice*Wildfire PM_{2.5} and the human larynx: paired in-event and clean-air assessment in one adult*

A single adult assessed twice — during a wildfire episode with PM_{2.5} reaching **117 µg m⁻³** and again at **<10 µg m⁻³** — with acoustic and aerodynamic voice measures, laryngeal endoscopy with stroboscopy, and *ventricular fold biopsies at both time points* analysed by qPCR for extracellular matrix and inflammatory markers. Speaking fundamental frequency fell relative to baseline, endoscopy showed visibly increased laryngeal mucus, and **MMP-1 expression roughly doubled**; other ECM and inflammatory genes moved inconsistently. *n* = 1. There is no control participant, no adjustment for phonatory load, hydration, intercurrent infection or the ordinary diurnal variation in fundamental frequency, and a two-point comparison cannot separate exposure from time. The reason to carry it is the design, not the result: paired human airway tissue across a natural exposure contrast is rare, and it establishes feasibility for a properly powered version. Treat the MMP-1 figure as a hypothesis.

doi: 10.1016/j.jvoice.2026.07.048

Hatem et al. 2026 Eur J Pediatr*Child characteristics, household and home environmental factors associated with eczema*

1,227 children aged 5–11 from 20 primary schools in Porto (46.5% response), analysed with mixed-effects logistic regression with a school random intercept and multiple imputation. Independently associated with higher odds: **residence within 300 m of a traffic zone, aOR 1.32 (1.11–1.48); postnatal secondhand smoke, aOR 1.48 (1.26–1.73)**; family history 3.53; higher family income 1.40; all-purpose cleaner use 1.16. Lower odds: maternal education 0.68, detached housing 0.67, fur blanket use 0.85. Household dampness interacted with residence type and winter ventilation frequency. Cross-sectional and self-reported, so temporality is unavailable — but the traffic term surviving a socioeconomic adjustment set this dense is the citable part.

doi: 10.1007/s00431-026-07327-1

Kanno et al. 2026 J Expo Sci Environ Epidemiol*Short-term PM_{2.5} and COVID-19 pneumonia in New York City, amplified by pre-existing cardiovascular disease*

15,361 NYC COVID-19 hospitalisations (March–December 2020) with short-term PM_{2.5} defined as the 7-day residential ZIP-code average from EPA CMAQ, and 15 ICD-coded circulatory conditions as the modifier. Modified Poisson regression for ARDS, pneumonia, ventilation and dialysis; Cox models for fatality and length of stay; RERI for additive interaction; TMLE for average treatment effects. **PM_{2.5} alone was null (RR 1.03, 95% CI 0.95–1.12) and prior myocardial infarction alone 1.34 (1.14–1.58), but jointly 1.67 (1.41–1.99)**; upper-quartile PM_{2.5} raised pneumonia risk only among those with cardiovascular conditions (**ATE 0.05, 0.02–0.08**). The interaction is the finding and it is the sort a main-effect analysis destroys. Weaknesses are real: ZIP-code CMAQ is a coarse surrogate for personal exposure and its error is spatially structured with deprivation, the effect appears only from July onward with no mechanism offered for the seasonal split, and the cohort is hospitalised patients only, so selection on admission is unaddressed. For network design the implication is concrete — the susceptible stratum is identifiable in advance, which is an argument for siting density by cardiovascular prevalence rather than by population alone.

doi: 10.1038/s41370-026-00957-5

Rice & Miller 2026 Pediatr Pulmonol*Air pollution, microbes and respiratory infection risk in children*

Narrative review joining two literatures usually kept apart: pollutant-driven epithelial barrier loss, oxidative stress and impaired innate/adaptive immunity, and the developing airway microbiome as a determinant of colonisation resistance. Argues that dysbiosis (antibiotics, air pollution, other perturbations) and pollutant exposure are partly substitutable routes to the same susceptibility phenotype, and notes that exposure to diverse environmental microbes — certain fungi included — may be protective. No quantitative synthesis and no bias assessment; useful as a framing citation for why an indoor sensing deployment in a childcare setting should log more than mass.

doi: 10.1002/ppul.71798

Tornevi et al. 2026 BMJ Open Respir Res

Short-term urban-background PM and the timing of confirmed COVID-19, three Swedish cities

Case-crossover on 3,999 PCR-confirmed cases from a prospective adult cohort in Stockholm, Gothenburg and Uppsala, 7 Mar 2020 – 26 Jan 2022, with daily PM_{2.5}, PM₁₀ and NO₂ from urban-background stations and distributed-lag (and non-linear) models adjusted for meteorology. **PM_{2.5} at lag 0: OR 1.084 (1.011–1.162) per IQR 3.3 µg/m³**, consistent across all three cities; PM₁₀ most consistent at lag 3; NO₂ lag 0 only after dropping the final wave. The authors themselves read the same-day lag as implausible for infection and propose symptom exacerbation prompting testing — i.e. the outcome is a *test date*, not an infection date, and the estimate is at least partly a health-seeking-behaviour effect.

doi: 10.1136/bmjresp-2025-003279

Wood et al. 2026 Lancet Public Health

Ultra Low Emission Zone and lung function growth in children: prospective parallel cohort

The strongest design in the issue. **3,414 children** recruited across 84 schools (44 London, 40 Luton) between June 2018 and April 2019 — *before* ULEZ implementation — with **3,209 (94.0%) in the FEV₁ analysis**. Luton functions as a contemporaneous control city, converting the policy rollout into a difference-in-differences natural experiment rather than a before–after series. Baseline London deficit –38 mL (95% CI –58 to –18) against a baseline NO₂ excess of **18.95 µg/m³**; over follow-up London FEV₁ growth exceeded Luton by **10 mL/year (5–15, p = 0.0002)** (233 vs 223 mL/year) and mean FEV₁ reached parity at four years. Clinically impaired lung function fell 14%→9% in London and 9%→7% in Luton. Two constraints on reading it as a PM result: the exposure contrast is expressed in NO₂, the exhaust marker, so PM and NO₂ effects are not separable; and two cities differ in more than their air, which parallel-cohort adjustment mitigates but does not eliminate.

doi: 10.1016/S2468-2667(26)00163-5

Yang et al. 2026a J Asthma

Atmospheric humidity and air pollutants versus self-reported chronic lung disease

16,204 CHARLS adults aged 45+, 2011/2013/2015 waves, six pollutants and six humidity indices at 1 km. Each IQR increase in pollutant concentration carried **18–45% higher odds** of self-reported CLD; PM₁₀ alone showed an approximately U-shaped exposure–response, and relative humidity an inverted-U, with a significant RH×sex interaction. Weighted quantile sum models were positive for both the pollutant and the humidity mixture. Self-reported outcome and a wide odds range quoted without per-pollutant intervals limit what can be taken from the point estimates; the reason to read it is the humidity treatment — six distinct indices rather than RH alone, which is the same distinction that separates a well-corrected from a badly-corrected low-cost nephelometer.

doi: 10.1080/02770903.2026.2721168

Zhang S et al. 2026 Environ Sci Technol

Respiratory and cardiovascular effects of ozone modified by PM_{2.5}

Two complementary designs testing one hypothesis: that 30-day mean PM_{2.5} *modifies* the acute cardiorespiratory response to ozone, and that unmodelled modification is part of why the ozone literature is inconsistent. In a healthy-adult panel, short-term ozone was associated with significantly larger decrements in FEV₁, FEF_{25–75} and PEF, and larger rises in systolic and diastolic blood pressure, under **PM_{2.5} above 22 µg/m³**. A meta-regression across published studies agreed: study areas with higher annual PM_{2.5} showed larger ozone effects on FEV₁ and blood pressure. **The abstract reports direction and significance but no interval estimates**, which is why this record contributes nothing to the forest plot. Panel size is not stated in the abstract either. Mechanistically the claim is priming rather than co-exposure confounding, and the meta-regression cannot separate the two — areas with high PM_{2.5} differ in many ways. Still the day’s one health record, and the modification framing is directly relevant to designing multi-pollutant sensor nodes.

doi: 10.1021/acs.est.6c05604

Reproductive & developmental

4 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (4 OF 4 RECORDS IN THIS SUBTOPIC)

Han et al. 2026 Environ Pollut

Aerosol pH associated with preterm birth risk: unmasking the hidden role of atmospheric acidity in maternal-fetal health

Metadata only. Surfaced through the Crossref by-ISSN sweep with no abstract deposited; title, journal, authorship and DOI verified, nothing else extracted. Worth flagging by title alone: aerosol acidity as an exposure metric in a reproductive endpoint is not something this watch has seen before, and it pairs directly with Chai et al.’s pH-governed partitioning result in yesterday’s issue. Queued for abstract retry.

doi: 10.1016/j.envpol.2026.129011

Jang et al. 2026 J Clin Endocrinol Metab

Ambient air pollution and age at natural menopause in a multi-ethnic US cohort

SWAN, **1,457 midlife women** across six US sites 1999–2015, five racial/ethnic groups, with 1 km² annual PM_{2.5}, NO₂ and O₃ linked to residence and age at natural menopause modelled as time-to-event by accelerated failure time. **NO₂ TR 0.983 (0.971–0.995), O₃ TR 0.983 (0.971–0.996)** for the highest versus lowest exposure quartile — roughly one year earlier menopause, which the authors benchmark against the 0.8-year advance seen in current smokers. **PM_{2.5} was not significantly associated**, which is the finding this watch should record plainly rather than fold into a general air-pollution claim. Residential address as the exposure surrogate over a 16-year window in a mobile midlife population is the main measurement limitation.

doi: 10.1210/clinem/dgag340

Wu et al. 2026 Ecotoxical Environ Saf

Prenatal PM_{2.5} constituents and early childhood obesity: organic matter dominates the mixture

339,435 mother–child pairs in Liaoning Province with prenatal PM_{2.5} and five constituents (organic matter, black carbon, ammonium, nitrate, sulfate) assigned from high-resolution atmospheric reanalysis, and overweight/obesity ascertained across the first two years. Single-pollutant models give **PM_{2.5} mass OR 1.050 (95% CI 1.036–1.064) per IQR**, with **OM 1.058 (1.043–1.072), NH₄⁺ 1.048 (1.035–1.062), NO₃[–] 1.048 (1.033–1.064) and BC 1.033 (1.020–1.047)**; sulfate is null. Quantile g-computation puts the mixture at **1.063 (1.050–1.076)** per quartile increase with **OM contributing 54.1%, NH₄⁺ 34.4% and NO₃[–] 11.5%**. The sample size makes the intervals tight, but tight intervals on reanalysis-assigned exposure are a precision claim about the model, not the exposure: constituent fields are far more uncertain than total mass and their errors are correlated with each other, which is exactly the condition under which QGC weights become unstable. Residential address at delivery stands in for the whole gestation, and no gestational windows are resolved. The direction is credible; the 54% attribution to OM is not something the exposure product can really support.

doi: 10.1016/j.ecoenv.2026.120670

Zhao et al. 2026 Chin Med J (Engl)*PM_{2.5} constituents and spontaneous abortion, modified by residential greenness*

20,117 pregnancies in Nanjing with residential PM_{2.5} speciated into NO₃⁻, SO₄²⁻, NH₄⁺, BC and OM; logistic regression with propensity-score matching plus two-pollutant models, WQS and BKMR for the joint effect. Per IQR: total mass **OR 1.18 (1.11–1.26)**, ammonium **1.19 (1.11–1.26)** highest, BC and OM lowest at 1.14 — and OM and BC alone lose significance once season of conception is added. Greenness (NDVI-1000 m) interacts significantly but non-monotonically (Q1 1.30, Q2 1.02, Q3 1.29, Q4 1.11), which undercuts the attenuation reading. Depression and anxiety scores predicted SAB independently with no interaction. Constituent correlations are high, so the ordering is suggestive rather than separable.

doi: 10.1097/CM9.0000000000004259

Mechanistic toxicology**8 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 8 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026a J Chin Med Assoc*PM_{2.5} suppresses IL-8-mediated epithelial migration in conjunctival cells*

Immortalised human conjunctival epithelial line exposed to PM_{2.5} with viability, apoptosis, scratch-migration and cytokine-array readouts. PM_{2.5} reduced viability time- and dose-dependently with only mild apoptosis, but markedly impaired migration and wound closure. Cytokine profiling showed *selective* remodelling rather than a general inflammatory response, with IL-8 the most strongly suppressed mediator; recombinant IL-8 rescued migration and partially rescued viability, and IL-8 neutralisation impaired migration at baseline. Single immortalised line, suspension dosing, no dose anchored to an ambient concentration — so the mechanism is credible and the dosimetry is not. The ocular surface remains an under-instrumented exposure site.

doi: 10.1097/JCMA.0000000000001418

Introna et al. 2026 ACS EST Air*Novel approach to assess the toxicity of PM_{2.5} on a subway platform using a mobile air-liquid interface system and submerged exposure*

A mobile ALI exposure system operated on the platform itself, with A549 monoculture and A549 + dTHP-1 coculture, benchmarked against conventional submerged laboratory exposure to the same metal-rich mechanical-wear particles. At the realistic in-situ ALI dose of **0.12 ± 0.07 µg/cm² there was no change in viability or inflammatory markers at 24 h**. Submerged at higher dose, viability fell in both cell types; A549 released IL-6 and TNFα at 100 µg/mL, and dTHP-1 released IL-8 at the lowest dose tested (10 µg/mL) with all four cytokines elevated at 100. The authors' own reading is the right one — longer in-situ exposures are needed to reach comparable dose — and the wider implication is that the submerged literature's dose range is not the ambient one.

doi: 10.1021/acsestair.6c00018

Jin & Zhang 2026 Reprod Toxicol*Astragaloside IV protects against PM_{2.5}-induced spermatocyte injury via cGAS-STING/Nrf2/HMOX1*

Network pharmacology and GEO reanalysis (GSE189187) to nominate targets, molecular docking, then validation in GC-2 spermatocytes. PM_{2.5} at **200 µg/mL** raised ROS 5.35-fold and apoptosis to 42.83%; AS-IV co-treatment restored viability to 83.3%, suppressed cGAS/p-TBK1/p-IRF3 and activated Nrf2/HMOX1, reducing apoptosis to 18.27%. The docking affinities exceed the reference inhibitors G150 and H-151, which is presented as validation and is not — docking scores do not rank cellular potency. The exposure is a cell-culture suspension four orders of magnitude above any inhaled concentration, and **AUC = 1.000** for a single gene in a public transcriptomic dataset is a small-sample artefact rather than a discriminative result. Carried at tier C for the pathway hypothesis only.

doi: 10.1016/j.reprotox.2026.109328

Li et al. 2026 FASEB J*Size-resolved carbon-black nanoparticle effects on the maternal–fetal lung axis*

Pregnant mice exposed to 30 nm or 120 nm carbon black under an explicitly high-dose mechanistic protocol, with H&E/Masson histology, dark-field hyperspectral imaging for particle localisation and chemical-isotope-labelling LC-MS metabolomics. Both sizes produced maternal inflammatory infiltration and fibrosis, **120 nm worse than 30 nm**; transplacental transfer was imaged directly in fetal lung. Fetal lungs showed no histological injury but raised oxidative and inflammatory markers, with size-specific metabolic signatures — 30 nm on histidine metabolism, 120 nm on butanoate and arachidonic acid. Two problems: the dose is not environmentally referenced, and the larger particle causing more damage inverts the surface-area expectation without a number- versus mass-dose accounting that would explain it.

doi: 10.1096/fj.202600065RR

Li et al. 2026b Environ Sci Technol*Component-resolved PM_{2.5} translocation across a human blood–brain-barrier immune platform*

Microfluidic BBB-immune microenvironment platform with compartment-resolved quantification of which PM_{2.5} constituents actually cross, rather than inference from downstream effect. High doses increased barrier permeability and drove constituent translocation; translocated material produced astrocyte reactivity, microglial M1 polarisation, elevated ROS, neuroinflammation and lipid-metabolic dysregulation. Component-resolved correlation implicates **chromium** and the water-soluble organic matter fraction — specifically the condensed species C₂₈H₇O₂N and C₂₁H₁₆N₂SP₂. The dose is described only as “high”, with no concentration given in the abstract and therefore no way to judge ambient relevance, which is the standard failure mode of this literature. The compartment-resolved measurement of translocation is nonetheless a genuine methodological advance over effect-only designs.

doi: 10.1021/acs.est.6c01030

Masci et al. 2026 Environ Toxicol Pharmacol*Air pollution and undernutrition induce rat alveolar macrophage functional dysregulation and compromise their response to a viral challenge*

Male Wistar rats, control vs. nutritional growth retardation (20% caloric restriction, 4 weeks), each with intranasal residual oil fly ash or vehicle; isolated alveolar macrophages assayed for viability, TNFα, oxidative stress, Nrf2 and LC3, then challenged with Poly(I:C). Undernutrition alone reproduced what ROFA did to well-nourished animals. **ROFA added little or nothing on top of undernutrition** — an attenuated rather than additive response — and after Poly(I:C), control macrophages mounted an inflammatory response that prior ROFA *amplified*, while undernourished macrophages failed to respond at all regardless of exposure. The interaction structure, not the main effect, is the finding: in a population that is already nutritionally compromised, the marginal immunological cost of PM may be small because the defence is already gone.

doi: 10.1016/j.etap.2026.105138

Sivakumar & Kurian 2026 Environ Toxicol*Duration-dependent transition from mitochondrial adaptation to persistent cardiac toxicity*

Female Wistar rats, whole-body PM_{2.5} at 250 µg/m³, 3 h/day for 1, 7, 14 or 21 days, with cardiac mitochondrial bioenergetics, redox balance, quality-control signalling and ex vivo function assessed after each duration and after a 24 h washout. **1–7 days: adaptive and fully reversible. ≥14 days: suppressed respiration and ETC complex activity, mtDNA depletion, impaired biogenesis, intramitochondrial metal accumulation — none recovered within the washout.** Concentration was held constant, so duration alone moved the system across the threshold. The 24 h washout is short, so “persistent” means “not recovered in 24 h”, which the authors state plainly; a longer recovery arm would settle it. The instrumentation implication is direct: a cumulative-duration-above-threshold metric is testable with continuous logging and invisible to a 24 h filter sample.

doi: 10.1002/tox.70189

Yan et al. 2026 Front Pharmacol*PM_{2.5} induces lung injury via mtDNA-cGAS-STING-mediated macrophage M1 polarization*

21-day subacute PM_{2.5} mouse model plus bone-marrow-derived macrophages in vitro. Exposure produced pulmonary inflammation and tissue injury with macrophage accumulation and M1 polarisation; mechanistically, mitochondrial damage released mtDNA and activated cGAS–STING. The load-bearing experiment is the rescue: **pharmacological cGAS inhibition with RU.521 attenuated STING activation, M1 polarisation and lung injury in vivo**, which converts a correlative pathway into an interventional one. Subacute rather than chronic, and a single exposure route — the dosimetry is the usual weak point.

doi: 10.3389/fphar.2026.1902928

Occupational & indoor**4 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (4 OF 4 RECORDS IN THIS SUBTOPIC)

Liu & Hu 2026 (preprint) Preprints.org*Multivariable determinants of indoor PM_{2.5} and infiltration dynamics in 48 residential buildings across eight cities of Northern China*

Paired indoor/outdoor PM_{2.5} at 5-minute resolution for seven consecutive days in each of four seasons across Beijing, Tianjin, Shijiazhuang, Taiyuan, Jinan, Zhengzhou, Xi'an and Anyang — **32,256 hourly observations from 48 residences**, with building characteristics, occupant behaviour and meteorology recorded alongside. A multivariable OLS model with cluster-robust standard errors ($R^2 = 0.879$) gives outdoor PM_{2.5} $\beta = 0.891$, window-open fraction $\beta = 0.572$, cooking $\beta = 0.049$ and **air-purifier operation** $\beta = -0.935$, all $p < 0.001$. Infiltration factor **0.28 ± 0.15**; outdoor sources contribute **57–69%** of indoor PM_{2.5}, peaking in the heating season; median deposition rate 0.53 h⁻¹ against a median air change rate of 0.52 h⁻¹, exceeding 2 h⁻¹ with windows open. Preprint, not peer reviewed — but the sample size and the seasonal design are both well beyond what this literature usually carries.

doi: 10.20944/preprints202608.1362.v1

Park et al. 2026b Environ Pollut*Phenotypic resistance, not community turnover, tracks antimicrobial use in swine-farm bioaerosol*

962 isolates across six species, six Korean swine farms stratified by defined daily dose for animals, four compartments (faecal, aerosol, pig nasal, worker nasal) and two seasons, with flow-cytometric viability and qPCR-corrected abundance feeding a Monte Carlo QMRA. *S. aureus* isolate-level resistance was higher on high-usage farms; the other four species and the community composition showed no usage association. *Escherichia* abundance fell ~four orders of magnitude on aerosolisation while *Staphylococcus* persisted and peaked in worker nasal samples. Median daily colonisation probability 1.98–8.63 × 10⁻⁴; the model was most sensitive to aerosol *genus composition*, not total bioaerosol load. Six farms cannot resolve a farm-level effect and the authors say so. For sensing: total particle or total bioaerosol load is the wrong exposure variable here.

doi: 10.1016/j.envpol.2026.128991

Xia et al. 2026 Front Pediatr*Enhanced filtration for indoor air quality in child-serving facilities*

Week-long baseline and follow-up measurements in 33 spaces across 12 facilities spanning daycare to high school, with staff surveys, as part of a community-based participatory partnership. Air change rates were low throughout (0.9–1.4 h⁻¹); baseline school-hours indoor PM_{2.5} 2.2–6.3 µg/m³ and PM₁₀ 5.7–26.9 µg/m³, highest in standard schools. Enhanced filtration cut PM_{2.5} **45–54%** in standard schools and the nursery and had little effect in childcare houses; a linear mixed model attributes ~51% of the PM_{2.5} reduction to the portable air cleaners. **Single-zone mass-balance prediction held for regular classrooms and the nursery but failed for portable buildings, very large rooms and repurposed houses.** That failure boundary is the most transferable result in the paper: it maps exactly onto where a deployed sensor earns its keep. No health endpoint was measured.

doi: 10.3389/fped.2026.1875263

Zhao et al. 2026 Indoor Air*CFD-based prediction of air change rate and PM in a dynamically occupied operating room*

Class 10,000 operating room in Dalian (Chinese GB 50333-2013), combining field measurement with CFD across 45 orthogonal scenarios, Sobol sensitivity, and three optimisation routes (Bayesian, multi-objective, response surface) to calibrate a non-uniformity coefficient and fit ACH–occupancy–PM. Response surface gave the lowest prediction error (**RMSE 0.032, MAE 0.024**); multi-objective optimisation was the most stable across runs (C_v 0.005). The premise is sound — fixed as-built ACH standards ignore occupancy dynamics — but the model is calibrated on one room in one climate, and the PM metric is cleanroom particle count rather than a health-relevant mass fraction.

doi: 10.1155/ina/5838379

Burden, policy & mitigation**9 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (9 OF 9 RECORDS IN THIS SUBTOPIC)

Alamsyah et al. 2026 Environ Sci Pollut Res Int

Industrial intensity and traffic congestion propagate air-pollution health vulnerability across district borders in Java

A health vulnerability index (HVI) for Java built by PCA over satellite pollutant fields, official statistics and OpenStreetMap congestion data, then regressed on industrial activity intensity and congestion in a spatial Durbin model to separate direct from spillover effects. **HVI ranges 29.2–62.3 on a 0–100 scale**, highest in West Java, Jakarta and Banten and lowest in Central Java, Yogyakarta and East Java; the exposure dimension varies most (13.18) against adaptive capacity (9.40) and sensitivity (5.17), and both industrial intensity and congestion act through **direct and indirect (spillover) channels**. The spillover result is the useful one — it says district-level mitigation will leak, which is a real argument for regional rather than municipal air management. But the index is a construct: PCA weights are data-driven rather than health-anchored, no health outcome is used to validate the HVI against observed morbidity, and OpenStreetMap congestion is an availability-biased proxy that is denser precisely in the wealthier districts. Uncertainty and sensitivity analysis is reported, which is more than most index papers do.

doi: 10.1007/s11356-026-38148-2

Anastasopoulos et al. 2026 Atmos Environ

Mortality attributed to ambient air pollution by anthropogenic emission sector in Canada

METADATA ONLY — a comparative health-burden analysis run at multiple geographic scales with sectoral attribution. Nine authors, Environment and Climate Change Canada style. No abstract in the Crossref deposit and nothing indexed elsewhere on 20 August; the sector-resolved burden numbers, which are the point of the paper, are therefore not reportable yet. Flagged for a later issue.

doi: 10.1016/j.atmosenv.2026.122310

Bhatta et al. 2026 Sci Total Environ

Outdoor air purification technologies: efficiency, sustainability and health impact

PRISMA-ScR review, 266 records screened to 28 studies across five technology domains. Reported bests: smog towers at **97.8% PM_{2.5} and 98.4% PM₁₀ removal under optimal conditions**; solar-chimney hybrids within 2–4% of dedicated systems while generating electricity; TiO₂-modified cementitious composites potentially carbon-negative over their lifecycle; a roadside hedge cutting leaf-surface PM_{2.5} magnetic loading 82%. The health-economics figure — 93,200 deaths averted in China at a 35 µg/m³ indoor target, positive net monetary benefit in 93% of cities — is modelled from indoor air cleaners used as an analogue for outdoor systems, which the authors flag and which should be read as a bound on plausibility, not an estimate. “Optimal conditions” removal efficiencies are chamber or near-field numbers and say nothing about neighbourhood-scale concentration. The review’s own priority list — standardised performance metrics, long-term field validation — is the honest summary.

doi: 10.1016/j.scitotenv.2026.182205

Bosso et al. 2026 J Environ Manage

AIRTREE v2.0: predicting ecosystem services provided by Italian urban parks under climate change scenarios

Multi-layer canopy model over 43 national-reforestation-programme sites under RCP4.5 and RCP8.5 to 2054, calibrated against five ICOS flux sites (Kling-Gupta efficiency up to 0.90). Under RCP8.5 net primary productivity rises 9% and evapotranspiration 8%, O₃ and NO₂ removal fall, and **PM₁₀ removal increases slightly**. Species responses diverge sharply — *Pinus halepensis* leads on PM removal, *Sorbus torminalis* on O₃ uptake, *Quercus pubescens* on carbon. Open-access tool; the species-resolved PM output is the part worth reusing.

doi: 10.1016/j.jenvman.2026.130738

Fares et al. 2026 Environ Pollut

Can urban green infrastructure influence the urban heat island effect and air quality nexus?

Review arguing that the thermal and air-quality benefits of green infrastructure are almost never assessed together, and synthesising the interaction: shading and evapotranspiration lower leaf surface temperature, which promotes thermophoretic deposition of particles, alongside foliar uptake of gaseous pollutants and inhibition of secondary aerosol formation. Well-maintained greenspace outperforms engineered installations, whose effects are more localised. The limitations section is the honest part — BVOC emission, small effect sizes, species and climate-zone dependence, and canopy-induced night-time warming.

doi: 10.1016/j.envpol.2026.128960

Liao et al. 2026 Environ Res

Beyond mortality: multi-outcome assessment of health economic benefits from air pollution control in China

Daily mortality, hospital admissions and outpatient visits across 18 Guangdong cities, 2015–2022, with generalized additive concentration–response models and valuation by value of statistical life plus direct medical cost. PM_{2.5} and NO₂ control delivered **10,305.35 and 1,017.00 million CNY** respectively; O₃ control returned *negative* net benefits on every outcome. The headline methodological number is that multi-outcome benefits of PM_{2.5} control **exceed mortality-only estimates by 31.79%** — i.e. the standard mortality-only accounting understates the case by roughly a third. Over 85% of mortality-related benefit accrues to the elderly.

doi: 10.1016/j.envres.2026.125501

Liu et al. 2026 Environ Sci Technol

Disproportionate effects of direct air capture on the regional health cobenefits of net-zero

Grid-level (36 km²) air-pollution mortality assessment along China’s net-zero CO₂ pathway 2020–2060, contrasting scenarios with and without large-scale DACCS deployment. DACCS substitutes for fossil-fuel phase-out and therefore forgoes the co-pollutant reduction that would otherwise occur: **1.75 million additional cumulative premature deaths**, with the losses concentrated in developing provinces (**Henan –0.13 million and Hebei –0.12 million avoided deaths**) via more active emissions trading. The paradox the authors report honestly is that inter-regional mortality-rate inequality narrows while the distribution of forgone benefit widens. Scenario-model output over a 40-year horizon, so the intervals are structural-uncertainty bounded rather than statistical, and the result is a property of the assumed trading architecture as much as of the technology.

doi: 10.1021/acs.est.6c04728

Moghaddam et al. 2026 (preprint) Research Square

Reducing particulate matter emissions in landfills using novel dust suppressants

Wind-tunnel factorial on soil from the Meyami landfill in Mashhad: three mesh sizes, four suppressants (compost, vermicompost, bentonite, nano-clay bentonite), three doses, two exposure times, two wind speeds. Mesh 20 was optimal across all suppressants (>85% reduction in wind-erosion rate); **compost at 10% dose reached 86.52%** and bentonite at 20% reached 80%, both at 4 m/s. A bench result on a single soil — no field validation and no airborne concentration measured — but the compost finding is operationally cheap enough to be worth a field trial.

doi: 10.21203/rs.3.rs-10431018/v1

Xia et al. 2026 (preprint) arXiv

Surface PM_{2.5} mass and composition under stratospheric sulfate geoengineering

Uses the GLENS ensemble (CESM1, stratospheric SO₂ injection balancing RCP8.5 forcing) to ask what solar geoengineering does to surface PM_{2.5}. Global mean PM_{2.5} falls relative to RCP8.5, but the reduction is carried entirely by **less dust** (wetter soils, higher leaf-area index over deserts) and **less sea salt** (slower winds). Excluding those two, PM_{2.5} rises, driven by aerosol-phase secondary organic aerosol partitioning in a cooler atmosphere. Notably, the injected SO₂ does **not** raise surface sulfate. Single model, single ensemble, an RCP8.5 baseline now far from plausible, and SOA gas precursors are prescribed — so the SOA increase is partly an artefact of that setup, as the authors acknowledge. Preprint; not peer reviewed.

doi: 10.48550/arXiv.2608.14928

Other clinical endpoints

13 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (13 OF 13 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026b Atmos Environ

Air pollution and sarcopenia: meta-analysis and network toxicology

METADATA ONLY — no abstract deposited. Title indicates a meta-analysis paired with a network-toxicology arm. Pooled estimate, heterogeneity and included-study count are all unknown from the deposit; a meta-analysis carried without its numbers is a placeholder, recorded so it is not missed on a later sweep.

doi: 10.1016/j.atmosenv.2026.122312

Choi et al. 2026 Environ Res

Time-varying long-term PM_{2.5} and leukaemia from childhood into early adulthood

National Health Insurance Service–National Sample Cohort, 2002–2020: **384,606** people aged 0–18 at enrolment, **272 incident leukaemias** (83 diagnosed after age 18), annual PM_{2.5} assigned to all 226 inland districts from an ML ensemble, Cox models with time-varying exposure and confounders plus two-pollutant sensitivity against NO₂ and O₃. **HR 1.40 (1.04–1.90) per 5 µg/m³**, stable when censored at 18. The exposure contrast is the problem: cohort mean **29.6 ± 3.1 µg/m³**, so a 5 µg/m³ increment exceeds 1.5 SD of the between-district spread and the estimate extrapolates beyond most of its own support, on 272 events with district-level assignment and no residential-mobility handling.

doi: 10.1016/j.envres.2026.125497

Guerrero-Morales et al. 2026 Environ Toxicol Pharmacol

Spatial distribution of long-term ultrafine particle concentrations and systemic lupus erythematosus burden in central Mexico

Municipal-level ecological analysis: SLE mortality 1998–2023 and hospitalisations 2008–2024 from national databases against satellite-model UFP for 2010–2019, using rate ratios, Jenks natural breaks and principal component analysis. National SLE mortality 0.23–0.86 per 100,000; municipal mean UFP roughly **1,500–24,000 particles/cm³**. Higher UFP and higher hospitalisation burden co-occur in metropolitan municipalities; mortality is heterogeneous and the authors say it should be read descriptively. Ecological design, no individual exposure, explicitly hypothesis-generating — but it is one of very few UFP–autoimmunity datasets from a middle-income country.

doi: 10.1016/j.etap.2026.105143

Kang et al. 2026 Environ Pollut

Short-term ambient air pollution and acute osteoarthritis morbidity across subgroups

Time-stratified case-crossover over **3,697,314 OA-related hospital visits** (ICD-10 M15–M19) from the Korean NHIS-NSC 2002–2019, linked to national pollutant fields, with health-screening data enabling individual-level effect modification. Per IQR: NO₂ **+5.28% (5.02–5.54)**, CO **+2.88%**, SO₂ **+1.36%**, PM₁₀ **+0.90% (0.73–1.07)**, O₃ **+0.76%**, PM_{2.5} **+0.73% (0.47–1.00)** — so the particulate terms are the weakest of the six and NO₂ is the only one robust in two-pollutant models. Enormous power makes intervals tight enough to exclude the null on effects this small, which is exactly when outcome-definition error matters: the authors state plainly that claims-based visit dates index healthcare utilisation, not symptom onset. A same-day traffic-pollution association with a discretionary outpatient visit is at least as consistent with behaviour as with pathophysiology.

doi: 10.1016/j.envpol.2026.128985

Lee et al. 2026 J Expo Sci Environ Epidemiol

Short-term ambient VOC mixtures, paediatric atopic dermatitis and asthma in Seoul

METADATA ONLY — short-term effects and attributable burden of ambient volatile organic compound mixtures on two paediatric outcomes. Included at the edge of scope: the exposure is VOCs rather than particles, but mixture-burden methods on a shared urban monitoring backbone are directly transferable, and the same Seoul network carries PM_{2.5}. No abstract available on 20 August, so neither the mixture method nor the burden fraction is reportable.

doi: 10.1038/s41370-026-00965-5

Lei et al. 2026 Atmos Environ

Hospital admission risk and burden of acute appendicitis attributable to PM_{2.5}

METADATA ONLY — no abstract deposited. Title indicates nationwide Chinese evidence on PM_{2.5} and acute appendicitis admissions with an attributable-burden component. No effect size, lag structure or exposure model is asserted here.

doi: 10.1016/j.atmosenv.2026.122311

Park et al. 2026a Gut Liver

Short-term PM_{2.5} and acute hepatic decompensation in cirrhosis

Nationwide time-stratified case-crossover on the Korean NHIS database, 2006–2021: 29,167 cirrhosis patients presenting to emergency departments with ascites, variceal bleeding, hepatic encephalopathy, jaundice or hepatorenal syndrome; control days matched on month and weekday; daily PM_{2.5} assigned on a 1 km residential grid; conditional logistic regression adjusted for temperature, humidity, holidays and weekday. **aOR 1.02 (1.00–1.04) per IQR increase**, with the top exposure quintile 5% above the bottom and *attenuation* at longer averaging windows. No subgroup interaction reached significance. The design controls time-invariant confounding well, but the lower bound sits on the null and the effect is small enough that residual exposure misclassification from grid-assigned concentration could account for it. A large, clean, honestly reported near-null.

doi: 10.5009/gnl260039

Poursafa & Jylhävä 2026 Ageing Res Rev*Air pollution and biological aging: mechanisms, biomarkers and reversibility*

Review organised by biomarker class rather than by outcome, which is what makes it useful. Evidence is graded as strongest for *system-level composites* — Klemmra–Doubal biological age, PhenoAge, frailty indices — and weakest, at meta-analytic level, for *epigenetic clocks*, which the authors attribute to heterogeneity in exposure assessment, design and target-tissue cell composition rather than to absent biology. Causal mediation across independent cohorts supports biological aging as a pathway from exposure to cardiovascular, neuropsychiatric and renal disease. Pollutants of concern are PM_{2.5}, PM_{0.1}, traffic-related pollution and household combustion. Named gaps: household air pollution and low-income settings are under-represented, and longitudinal multi-omic designs are scarce. Air-quality and clean-cooking improvements are reported to partially reverse the acceleration — the strongest claim in the paper and the one most dependent on the intervention literature it summarises.

doi: [10.1016/j.arr.2026.103297](https://doi.org/10.1016/j.arr.2026.103297)**Wang et al. 2026** Ecotoxical Environ Saf*Early-life PM_{2.5} mass and components and childhood anaemia across China, India and Africa*

Cross-sectional analysis of **449,894 children under five** from China, India and 26 African countries, with prenatal, postnatal and cumulative PM_{2.5} and five components (black carbon, organic carbon, sulfate, dust, sea salt) from NASA MERRA-2 reanalysis. Anaemia prevalence is **26.9% in China, 62.6% in India, 64.5% in Africa**. Per 1 $\mu\text{g m}^{-3}$ of cumulative PM_{2.5}: **OR 1.019 (95% CI 1.009–1.028) in China, 1.009 (1.008–1.009) in India, 1.001 (1.000–1.002) in Africa**, with organic carbon and dust positive in all three. Population attributable fractions are **28.46%, 10.64% and 0.69%**. The regional ordering of effect size and PAF runs *opposite* to the ordering of anaemia burden, and the paper does not resolve this: the most economical explanation is that MERRA-2 at $0.5^\circ \times 0.625^\circ$ has little within-country exposure contrast in Africa and correspondingly attenuated coefficients, not that African children are less susceptible. Cross-sectional design cannot order exposure and outcome, and anaemia in these settings is dominated by iron deficiency, malaria and haemoglobinopathy — confounders that co-vary strongly with dust exposure. Read the component ranking, not the PAFs.

doi: [10.1016/j.ecoenv.2026.120658](https://doi.org/10.1016/j.ecoenv.2026.120658)**Wang et al. 2026** Atmos Environ*Long-term PM_{2.5}, aging hallmarks and age-related disease*

METADATA ONLY — no abstract deposited. Title indicates a nationwide prospective cohort linking long-term PM_{2.5} to aging hallmarks and age-related disease incidence. Cross-reads directly with Poursafa & Jylhävä above, which is the reason to retrieve it in full rather than wait for indexing.

doi: [10.1016/j.atmosenv.2026.122313](https://doi.org/10.1016/j.atmosenv.2026.122313)**Wei et al. 2026** Environ Res*Chemical composition of fine particulate matter and all-cause mortality of older lung cancer patients*

528,127 SEER-Medicare adults aged ≥ 65 diagnosed with lung cancer 2000–2019, followed annually; two-year moving-average exposure to 15 PM_{2.5} components from high-resolution spatiotemporal models linked by residential ZIP code; generalized weighted quantile sum regression with random holdouts, adjusted for demographics, histology, stage, first-course treatment, comorbidity and neighbourhood covariates. **RR 1.011 (95% CI 1.010–1.013) per decile increase across all components**. Contribution differences were modest but silicon, nitrate, vanadium, zinc and iron led — road dust, traffic, fossil-fuel and heavy-fuel-oil combustion. Exploratory subgroups suggest a stronger association in non-small-cell, later-stage and female patients.

doi: [10.1016/j.envres.2026.125538](https://doi.org/10.1016/j.envres.2026.125538)**Zewdie et al. 2026** Cancer Epidemiol*Wildfire smoke PM_{2.5} and cancer history prevalence at census-tract level*

Daily wildfire-specific smoke PM_{2.5} aggregated to cumulative (2006–2020) and recent (2018–2020) means, linked to 2022 tract-level cancer history prevalence for all contiguous-US tracts; quasi-Poisson GAMs with random effects and spatial splines, adjusted for sociodemographics, behaviours, health status, density, climate and region. **No association in fully adjusted models for either exposure window**. Regional effect modification produced small positive estimates in the Midwest (**PR 1.020, 1.013–1.026**) and Northeast (**PR 1.040, 1.032–1.049**) — the two regions with the least smoke exposure, a spatial pattern more consistent with residual confounding than with dose. Cross-sectional and ecological, with prevalence of *ever* having cancer set against a 15-year exposure window that cannot precede diagnosis for most cases; latency alone makes this design close to uninformative for incidence. The value here is a well-powered, honestly reported null in a literature prone to amplification.

doi: [10.1016/j.canep.2026.103199](https://doi.org/10.1016/j.canep.2026.103199)**Zhang et al. 2026** Environ Sci Technol*PM_{2.5} exposure and the prognosis of severe fever with thrombocytopenia syndrome*

A cohort of **3,583 hospitalised SFTS patients** with long-term pre-admission ambient PM_{2.5}: **HR 1.16 (95% CI 1.04–1.30) per 10 $\mu\text{g}/\text{m}^3$** for SFTS-attributable death, with exploratory mediation implicating viral load. A counterfactual burden assessment estimates **137 avoidable SFTS deaths in Xinyang, 2010–2022** had annual PM_{2.5} met the Chinese national standard. The experimental arm shows PM_{2.5} pre-exposure increasing SFTSV replication and tissue injury; transcriptomics implicate mitochondrial redox and ROS pathways, and ROS/MDA measurement plus N-acetylcysteine intervention support oxidative priming as the upstream step. Emerging-infection prognosis is a genuinely new endpoint class for this watch.

doi: [10.1021/acs.est.6c03214](https://doi.org/10.1021/acs.est.6c03214)

Pooled corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Chen et al.	Integrated plasma proteomics and metabolomes reveal potential biomolecules linking PM _{2.5} to depression	<i>Environ Toxicol Pharmacol</i>	Cohort	PM _{2.5} + 5 constituents (NH ₄ , NO ₃ , SO ₄ , EC, OC)	UK
He et al.	Large-scale plasma proteomics identifies protein signatures linking air pollution to epilepsy	<i>Ecotoxical Environ Saf</i>	Prospective cohort	PM _{2.5} , PM _{coarse} , NO ₂ , NO from land-use regression	Not stated
Parmar et al.	Short-term PM _{2.5} and its chemical components and neuropsychiatric hospitalisations	<i>Environ Int</i>	Time-series	Ground-WRF-Chem fused PM _{2.5} at 5 km, speciated into BC, OC, OIN	India
Perera et al.	Declining prenatal air pollution and improving cognitive scores across successive NYC cohorts	<i>Environ Health</i>	Prospective cohort	Prenatal personal/modelled PM _{2.5} , NO ₂ and airborne PAH	USA
Cardiovascular & metabolic					
Chen et al.	Pollutant-specific plasma proteomic signatures, mediators and atrial-fibrillation-free life lost	<i>Ecotoxical Environ Saf</i>	Prospective cohort	Long-term modelled PM _{2.5} and PM ₁₀ (plus NO ₂ , NO _x , SO ₂ , benzene, O ₃)	UK
Liu et al.	Household clean cooking fuel transition and attenuated cardiopulmonary function decline: a national longitudinal study of middle-aged and older adults in China	<i>Indoor Air</i>	Cohort	Household solid-fuel to clean-fuel cooking transition	China
Nong et al.	Air pollution and diabetic microvascular complications: a prospective cohort study	<i>PLoS One</i>	Prospective cohort	PM ₁₀ , PM _{2.5} , NO ₂ (annual, quartiles)	UK
Reproductive & developmental					
Han et al.	Aerosol pH associated with preterm birth risk: unmasking the hidden role of atmospheric acidity in maternal-fetal health	<i>Environ Pollut</i>	Metadata only (no abstract)	Aerosol acidity (pH) (not extractable)	Not stated
Jang et al.	Ambient air pollution and age at natural menopause in a multi-ethnic US cohort	<i>J Clin Endocrinol Metab</i>	Prospective cohort	1 km ² modelled annual PM _{2.5} , NO ₂ and O ₃ at residence	USA
Wu et al.	Prenatal PM _{2.5} and its chemical constituents and early childhood overweight/obesity: organic matter dominates the mixture effect	<i>Ecotoxical Environ Saf</i>	Prospective cohort (mixtures)	Gestational PM _{2.5} mass + OM, BC, NH ₄ ⁺ , NO ₃ ⁻ , SO ₄ ²⁻ (reanalysis)	China
Zhao et al.	PM _{2.5} constituents and spontaneous abortion, modified by residential greenness	<i>Chin Med J (Engl)</i>	Prospective cohort (mixtures)	Residential annual PM _{2.5} with NO ₃ ⁻ , SO ₄ ²⁻ , NH ₄ ⁺ , BC and OM fractions	China
Respiratory & allergic					
Ali et al.	Symptom burden among adult out-patients during the 2024-2025 winter smog season in Lahore	<i>Public Health Chall</i>	Cross-sectional (multistage)	City-level PM _{2.5} (mean 96 µg/m ³), PM ₁₀ , CO, AQI	Pakistan
Erickson-DiRenzo et al.	Wildfire PM _{2.5} and the human larynx: paired in-event and clean-air voice, endoscopy and ventricular-fold biopsy in one adult	<i>J Voice</i>	Repeated-measures biomarker	Wildfire episode PM _{2.5} up to 117 µg/m ³ vs <10 µg/m ³ baseline	USA
Hatem et al.	Child characteristics, household, and home environmental factors associated with eczema: a cross-sectional study	<i>Eur J Pediatr</i>	Cross-sectional survey	Traffic-zone proximity (<300 m) and postnatal secondhand smoke	Porto, Portugal
Kannoth et al.	Short-term PM _{2.5} and COVID-19 pneumonia in New York City: risk amplified by pre-existing cardiovascular disease	<i>J Expo Sci Environ Epidemiol</i>	Retrospective cohort	7-day residential ZIP-code CMAQ PM _{2.5} prior to admission	USA
Rice & Miller	Air pollution, microbes and respiratory infection risk in children	<i>Pediatr Pulmonol</i>	Narrative review	PM _{2.5} and indoor combustion PM (qualitative)	Global

First author	Focus	Journal	Design	Metric	Region
Tornevi et al.	Short-term urban-background PM _{2.5} and the timing of confirmed COVID-19, three Swedish cities	<i>BMJ Open Respir Res</i>	Case-crossover	Daily urban-background PM _{2.5} and PM ₁₀ from fixed reference monitors	Sweden
Wood et al.	Ultra Low Emission Zone and lung function growth in children: parallel cohort study	<i>Lancet Public Health</i>	Prospective cohort	Modelled residential annual NO ₂ and PM (LAEI dispersion model)	UK
Yang et al.	Interactive associations of atmospheric humidity and air pollutants with self-reported chronic lung disease	<i>J Asthma</i>	Prospective cohort (mixtures)	PM _{2.5} , PM ₁₀ + 4 gases at 1 km, with 6 humidity indices	China
Zhang S et al.	Respiratory and cardiovascular effects of ozone modified by PM _{2.5}	<i>Environ Sci Technol</i>	Panel study	30-day mean PM _{2.5} as effect modifier of short-term O ₃	Guangzhou, China + literature
Mechanistic toxicology					
Chen et al.	PM _{2.5} suppresses IL-8-mediated epithelial migration of human conjunctival epithelial cells	<i>J Chin Med Assoc</i>	In vitro (cell lines)	PM _{2.5} (suspension, time- and dose-response)	Laboratory
Introna et al.	Novel approach to assess the toxicity of PM _{2.5} on a subway platform using a mobile air-liquid interface system and submerged exposure	<i>ACS EST Air</i>	In vitro exposure	Subway-platform PM _{2.5} , metal-rich; ALI dose 0.12 +/- 0.07 ug/cm ²	Sweden (Stockholm Univ. affiliation; platform city not stated)
Jin & Zhang	Astragaloside IV protects against PM _{2.5} -induced spermatocyte injury via cGAS-STING/Nrf2/HMOX1	<i>Reprod Toxicol</i>	In vitro exposure	PM _{2.5} suspension at 200 ug/mL in GC-2 spermatocytes	China
Li et al.	Size-resolved carbon-black nanoparticle effects on the maternal-fetal lung axis	<i>FASEB J</i>	Animal model	Ultrafine carbon-black nanoparticles, 30 nm and 120 nm, high-dose mechanistic exposure	Laboratory (murine)
Li et al.	Component-resolved PM _{2.5} translocation across a human blood-brain barrier immune platform	<i>Environ Sci Technol</i>	In vitro organ-on-chip	Ambient PM _{2.5} extract, compartment-resolved constituent quantification	China
Maschi et al.	Air pollution and undernutrition induce rat alveolar macrophage functional dysregulation and compromise their response to a viral challenge	<i>Environ Toxicol Pharmacol</i>	Animal experiment	Residual oil fly ash (ROFA), intranasal, sub-chronic	Argentina
Sivakumar & Kurian	Repeated PM _{2.5} inhalation drives a duration-dependent transition from mitochondrial adaptation to persistent cardiac toxicity	<i>Environ Toxicol</i>	Animal model	PM _{2.5} 250 µg/m ³ , 3 h/day, 1-21 days (whole-body inhalation)	Laboratory
Yan et al.	PM _{2.5} induces lung injury via mtDNA-cGAS-STING-mediated macrophage M1 polarization	<i>Front Pharmacol</i>	Animal + in vitro	PM _{2.5} , 21-day subacute exposure	Laboratory (mouse + BMDM)
Occupational & indoor					
Liu & Hu 2026	Multivariable determinants of indoor PM _{2.5} and infiltration dynamics in 48 residential buildings across eight cities of Northern China	<i>Preprints.org</i>	Measurement campaign	Paired indoor/outdoor PM _{2.5} ; infiltration factor F _{inf} 0.28 +/- 0.15	Northern China (8 cities)
Park et al.	Within-species phenotypic resistance rather than community turnover tracks antimicrobial usage in swine-farm bioaerosol	<i>Environ Pollut</i>	Environmental surveillance campaign	Viable bioaerosol abundance (flow cytometry + qPCR-corrected)	South Korea
Xia et al.	Effectiveness of enhanced filtration interventions for indoor air quality in child-serving facilities	<i>Front Pediatr</i>	Trial / intervention	Indoor PM _{2.5} and PM ₁₀ , week-long pre/post continuous	USA
Zhao et al.	CFD-based prediction of air change rate and particulate matter in a dynamically occupied operating room	<i>Indoor Air</i>	Physical exposure model	Indoor PM (Class 10,000 cleanroom)	Dalian, China
Sensing, forecasting & instrumentation					
Ayele et al.	Fine-scale variations in black carbon and TSP chemical composition at a heavy-traffic intersection, Addis Ababa	<i>Environ Sci Atmos</i>	Metadata only (no abstract)	Equivalent black carbon + total suspended particulate organics	Ethiopia

First author	Focus	Journal	Design	Metric	Region
Band & Mohammadi 2026	Shift-aware dual-encoder transfer learning for short-horizon PM _{2.5} forecasting in data-limited target domains	<i>arXiv (preprint)</i>	Deep-learning model	Hourly regulatory PM _{2.5} , 10 US source sites -> 77 Taiwan target stations	Taiwan
Clark & Bashar	How IoT enables the Protective Action Decision Model during a wildfire event	<i>Risk Anal</i>	Cross-sectional survey	Hyper-local real-time PM _{2.5} (IoT)	Oregon, USA
Dal Porto et al.	Humidified CAPS-PM_SSA with a revised truncation-correction methodology	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Aerosol optical extinction and scattering, single-scattering albedo	Chamber / laboratory
Diana et al.	Validated multi-technique workflow for inorganic characterisation of brake-wear PM ₁₀ and source-marker identification	<i>Talanta</i>	Bench / device experiment	WLTP brake-dynamometer PM ₁₀ , low-steel and NAO pads on cast-iron/coated discs	Italy
Fei et al.	SHAP-interpretable machine learning integrating exposures and multi-omics reveals immune alterations and biomarkers in COPD-lung cancer comorbidity	<i>Front Immunol</i>	Machine-learning model	Particulate-matter pollution as a GBD exposure risk factor	Global
Filioglou et al.	Particle linear depolarisation ratio at 1565 nm from Halo Doppler lidar: smoke vs volcanic ash	<i>Atmos Chem Phys</i>	Multi-instrument field evaluation	Lidar depolarisation ratio and aerosol layer optical properties at 1565 nm	Finland
Kalkavouras et al.	Long-term (10-year) trends in ultrafine particle number concentrations and size distributions in Athens	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Ultrafine particle number concentration + size distribution	Europe
Legaria-Santiago et al.	Leveraging remote traffic data for local air pollutant estimation: a scenario-based machine learning study across London monitoring sites	<i>Atmosphere</i>	Machine-learning model	NO ₂ , PM ₁₀ , PM _{2.5} , O ₃ from remote traffic counts	London, UK
Li et al.	Crowdsourced, community-engaged VOC monitoring and source apportionment near a landfill	<i>Environ Sci Technol</i>	Field measurement	Gas-phase BTEX by EPA TO-17, not PM	USA
Ng et al.	Impacts of droughts on biomass burning emissions, air quality, and public health in the Amazon	<i>Atmos Chem Phys</i>	Satellite + chemical transport model	Biomass-burning PM _{2.5} and CO (GFEDv4s / GEOS-Chem HP)	Amazon, Brazil
Perisic et al.	Explainable AI resolves ten recurrent environmental settings behind urban PM ₁₀ variability at a single regulatory station	<i>Sci Total Environ</i>	Algorithm development	Regulatory PM ₁₀ /PM _{2.5} + NO ₂ , SO ₂ , O ₃ , meteorology, 2020-2023	Europe
Rapp et al.	Curve fitting algorithm for multimodal particle size distributions: a theoretical basis	<i>Atmos Meas Tech</i>	Measurement campaign	Mobility particle size distributions, number-based	USA
Saravagna Sai Lakshmi et al.	CORTA-Net: hybrid deep learning with attention-based interpretability for PM _{2.5} forecasting	<i>J Vis Exp</i>	Machine-learning model	Hourly regulatory-station PM _{2.5} plus MODIS fire counts	India
Shlipak et al.	Spatial and temporal trends of ambient PM _{2.5} and source-apportioned black carbon in Addis Ababa	<i>ACS EST Air</i>	Network deployment + field evaluation	PM _{2.5} + black carbon (BC)	Addis Ababa, Ethiopia
Wang et al.	Remote calibration of a nationwide dense low-cost sensor network for hyperlocal air surveillance	<i>Environ Sci Technol</i>	Machine-learning model	Low-cost sensor PM _{2.5} vs regulatory reference monitors	USA
Wei et al.	Stacked Transformer-XGBoost model with SHAP interpretability for air quality index prediction	<i>iScience</i>	Machine-learning model	AQI driven by PM _{2.5} , PM ₁₀ , NO ₂	Linfen, China
Yang et al.	Dust transport and local anthropogenic emissions differently shape ice-nucleating particles in an industrial urban atmosphere	<i>Atmos Chem Phys</i>	Measurement campaign	Immersion-mode INP concentration, size distribution, PM _{2.5} PMF factors	China
Zeng et al.	Aerosol vertical distributions determine the seasonal relationships between column AOD and surface PM _{2.5} over Europe	<i>Atmos Environ</i>	Metadata only (no abstract)	Column AOD vs surface PM _{2.5} (not extractable)	Europe
Zhang et al.	Wearable device for direct assessment of particle deposition in the human respiratory system	<i>Environ Sci Technol</i>	Field measurement	Paired inhaled and exhaled size-resolved PM, PM ₁ -PM ₁₀	China

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Zhao et al.	Monte-Carlo inversion of aerosol complex refractive index from multi-wavelength lidar plus size distribution	<i>J Aerosol Sci</i>	Metadata only (no abstract)	Complex refractive index, multi-wavelength lidar backscatter, particle size distribution	Not stated
Zundel et al. 2026	Wearable personal PM _{2.5} monitoring during wildfire smoke episodes in Metro Detroit	<i>medRxiv</i>	Proxy validation vs personal exposure	Personal PM _{2.5} (AirBeam3) vs PurpleAir	Detroit, USA
Exposure assessment & modelling					
Abadi et al.	Future projections of surface dust concentrations in major dust source regions of the Mesopotamian plains	<i>Atmos Res</i>	Metadata only (no abstract)	Projected surface mineral dust (not extractable)	Iraq
Alaimo et al.	Elemental composition and Pb isotopic signatures in pine needles (<i>Pinus pinea</i> L.) from the industrialized Milazzo area	<i>Atmosphere</i>	Deposition sampling	Atmospheric trace-element deposition; Pb isotope source apportionment	Milazzo, Italy
Albugami	Dust event characteristics, transport pathways and source regions over Riyadh	<i>Atmosphere</i>	Field campaign / remote sensing	Coarse-mode dust (AOD, Angstrom exponent)	Riyadh, Saudi Arabia
Alshehri & El-Hamid 2026	Sentinel-5P and MODIS land-surface coupling of pollutant fields over Riyadh, 2018-2025	<i>Research Square (preprint)</i>	Spatial analysis / GIS	TROPOMI NO ₂ /SO ₂ /CO/O ₃ + Absorbing Aerosol Index; MODIS LST/NDVI	Middle East & N. Africa
Ashta et al.	Wet and dry atmospheric deposition of microplastics at urban, suburban, rural and mountainous sites	<i>Atmos Chem Phys</i>	Deposition sampling	Microplastics 20-215 µm	Switzerland
Awan et al.	BRIQ-PM _{2.5} : quantile-mapping attribution of industrial excess PM _{2.5} across Pakistan	<i>Aerosol Air Qual Res</i>	Spatial analysis / GIS	Satellite-derived surface PM _{2.5}	Pakistan
Blanco-Alegre et al.	Emissions from prescribed fires in two Mediterranean shrublands: chemical and morphological analysis	<i>Atmos Chem Phys</i>	Field measurement	Gravimetric PM _{2.5} emission factors + OC/EC, ions, SEM morphology	Spain
Cai et al.	Effect of ammonia on reactive oxygen species generation in secondary organic aerosols	<i>Atmos Environ</i>	Metadata only (no abstract)	SOA reactive oxygen species yield (not extractable)	Not stated
Chai et al.	Role of ammonia, aerosol pH and liquid water content in secondary inorganic aerosol formation in suburban Nanjing	<i>Environ Pollut</i>	Measurement campaign	PM _{2.5} secondary inorganic aerosol (mean 34.4 µg/m ³)	Nanjing, China
Chuang et al.	Simulating PM _{2.5} in southern Taiwan under high-pressure return flow during the Kao-Ping Experiment	<i>Atmos Res</i>	Metadata only (no abstract)	PM _{2.5} (not extractable)	Kao-Ping, Taiwan
Defrance 2026	Beyond the fire perimeter: the smoke footprint that constrains outdoor activity dwarfs the burned area of the July 2026 Gironde-Landes wildfire episode	<i>Research Square</i>	Modelling / inventory	CAMS analysis PM _{2.5} + 40 background monitors	Gironde-Landes, France
Dias et al.	Micro-scale spatial patterns of PM _{2.5} across urban green and non-green areas in an arid city	<i>Bull Environ Contam Toxicol</i>	Measurement campaign	Mobile-monitored PM _{2.5} along a fixed transect	Argentina
Do et al.	Field-constrained diurnal viscosity variability of urban PM _{2.5} and implications for N ₂ O ₅ uptake	<i>Environ Sci Technol</i>	Laboratory metrology	PM _{2.5} phase state / viscosity	Ansan, South Korea + 14 megacities
Dovrou et al.	Wintertime wildfires reshape urban atmospheric chemistry: evidence from the January 2025 Los Angeles fires	<i>Environ Sci Atmos</i>	Metadata only (no abstract)	Wildfire-influenced urban aerosol (not extractable)	USA
Duan et al.	Regime-dependent aerosol pH variability in ammonia-rich urban Beijing from explainable machine learning	<i>Atmos Chem Phys</i>	Machine-learning model	PM ₁ mass and composition + thermodynamic pH	China
Fan et al.	Multiple amides synergistically enhance sulfuric acid-ammonia nucleation by an order of magnitude over single-base systems	<i>iScience</i>	Chemical transport modelling	Modelled SA-NH ₃ -MPA/MAA cluster formation (new particle formation)	China
Goncalves et al.	Extreme Saharan dust outbreak: air quality crisis in NW Spain	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Saharan mineral dust (not extractable)	NW Spain
Guo et al.	Meteorology-driven long-range transport to an ultra-low emission region during the monsoon	<i>Atmos Environ</i>	Metadata only (no abstract)	Not extractable from registered metadata	China

First author	Focus	Journal	Design	Metric	Region
Ianiri et al.	Street- vs roof-level deposition fluxes of POPs and metals, normalised to sedimentable PM	<i>Environ Pollut</i>	Deposition sampling	Settleable particulate matter, monthly deposimeter collection	Italy
Jin et al.	Hourly 2 km Himawari-8 aerosol fine-mode fraction retrieval improves PM _{2.5} mapping	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	Satellite AOD fine-mode fraction to surface PM _{2.5}	China
Joshi et al. 2026	Environmental drivers of foliar particulate accumulation, leaf structural and biochemical responses in urban trees in Delhi	<i>Research Square</i>	Deposition sampling	Leaf-deposited coarse (10-100 µm) and fine (2.5-10 µm) PM	Delhi, India
Kan et al.	Short-term observations of airborne microplastics in Phnom Penh: concentrations, aerodynamic size distribution, and polymer composition	<i>Atmosphere</i>	Measurement campaign	Airborne microplastics in TSP and PM _{2.5} (uFTIR-ATR)	Phnom Penh, Cambodia
Kommula et al.	Comparable organic aerosol mass enhancement during photochemical and dark oxidation of biomass burning emissions	<i>Environ Sci Technol</i>	Chamber emission characterisation	PM ₁ organic aerosol / BB-SOA	Chamber (Finland)
Lazo et al.	Biomonitoring trace metal deposition with <i>Ligustrum lucidum</i> : Albanian fire incidents	<i>J Environ Qual</i>	Deposition sampling	Fire-derived trace-metal deposition on leaves (ICP-MS)	Elbasan, Albania
Liu et al.	Identification of dominant weather patterns and circulation regulation mechanisms of winter PM _{2.5} pollution in Iran	<i>Atmos Res</i>	Metadata only (no abstract)	Winter PM _{2.5} under synoptic weather types (not extractable)	Iran
Mahia et al.	Real-time effects of street geometry on elderly well-being under combined air-heat exposure	<i>J Environ Manage</i>	Observational exposure study	Personal portable PM ₁₀ (to 18.77 µg/m ³) and NO ₂ , wearable physiology	Italy
Moryani et al.	PM _{2.5} -bound Cu, Cd, Ni, Pb and Zn in 20 schools across two contrasting South China cities	<i>J Appl Toxicol</i>	Source apportionment	Heavy-metal content of PM _{2.5} , summer and winter	China
Naghizade et al.	Multi-class organic marker profiling of urban PM ₁₀ in Strasbourg	<i>Atmos Environ</i>	Metadata only (no abstract)	PM ₁₀ organic markers (not extractable)	Strasbourg, France
Qing et al.	Seasonal and urban-suburban characteristics, emission and spatial sources of PM _{2.5} -bound PAHs and n-alkanes in Wuhu City	<i>Atmos Pollut Res</i>	Metadata only (no abstract)	PM _{2.5} -bound PAHs and n-alkanes (not extractable)	Wuhu, China
Saarikoski et al.	Role of hydrocarbons and oxygenated organic species in traffic-derived aerosol	<i>Atmos Chem Phys</i>	Measurement campaign	Submicron organic aerosol (AMS)	Helsinki, Finland
Wang Q et al.	Quantitative analysis of aerosol acidity and its driving factors in the Guanzhong area	<i>Atmos Chem Phys</i>	Measurement campaign	PM _{2.5} , water-soluble inorganic ions, pH	Guanzhong Plain, China
Wang T et al.	Chlorine radical cycling on photoactive particulate matter amplifies photolytic NO _x release	<i>Environ Sci Technol</i>	Controlled chamber factorial	Photoactive PM surfaces, soot	Laboratory
Yang et al.	Impacts of anthropogenic aerosols on a snowfall event in the Guanzhong Basin and surrounding areas	<i>Atmos Chem Phys</i>	Modelling / inventory	Anthropogenic aerosol (ARI and ACI pathways)	Guanzhong Basin, China
Yu et al.	Probabilistic assessment of PM _{2.5} and ozone co-occurring extremes under wildfire smoke	<i>Environ Sci Technol</i>	Bayesian + geospatial regression	PM _{2.5} + O ₃ joint extremes	Contiguous USA
Zaman & Mim	Decadal variation of atmospheric PM _{2.5} over Bangladesh from MERRA-2 reanalysis	<i>Environ Sci Atmos</i>	Machine-learning model	MERRA-2 reanalysis PM _{2.5} and speciated components	Bangladesh
Zheng et al.	Aerosol modulation of precipitation systems migrating from mountains to plains in Northern China during early autumn	<i>Atmos Res</i>	Metadata only (no abstract)	Aerosol-precipitation interaction (not extractable)	Northern China
Burden, policy & mitigation					
Alamsyah et al.	Industrial intensity and traffic congestion propagate air-pollution health vulnerability across district borders in Java	<i>Environ Sci Pollut Res Int</i>	Ecological panel	Satellite pollutant fields + OpenStreetMap congestion, district level	Indonesia
Anastasopoulos et al.	Comparative health burden analysis at multiple geographic scales: mortality attributed to ambient air pollution by sector	<i>Atmos Environ</i>	Metadata only (no abstract)	Sectoral anthropogenic PM _{2.5} (not extractable)	Canada
Bhatta et al.	Outdoor air purification technologies for urban PM and gaseous pollutant mitigation: a comprehensive review	<i>Sci Total Environ</i>	Systematic review (PRISMA)	PM _{2.5} and PM ₁₀ removal efficiency across five technology domains	Global

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Bosso et al.	AIRTREE v2.0: predicting ecosystem services provided by Italian urban parks under climate change scenarios	<i>J Environ Manage</i>	Modelling / inventory	PM ₁₀ and PM _{2.5} removal by urban canopy	Italy (43 sites)
Fares et al.	Can urban green infrastructure influence the urban heat island effect and air quality nexus?	<i>Environ Pollut</i>	Review / synthesis	PM deposition and thermophoresis onto green infrastructure	Global
Liao et al.	Beyond mortality: a multi-outcome assessment of health economic benefits from air pollution control in China	<i>Environ Res</i>	Burden / economic model	PM _{2.5} , NO ₂ and O ₃ concentration-response	Guangdong, China (18 cities)
Liu et al.	Disproportionate effects of direct air capture on regional health cobenefits of net-zero	<i>Environ Sci Technol</i>	Health impact model	Grid-level (36 km ²) modelled PM _{2.5} and O ₃ attributable mortality	China
Moghaddam et al. 2026	Reducing particulate matter emissions in landfills using novel dust suppressants	<i>Research Square</i>	Chamber / laboratory	Wind-eroded fugitive dust (wind-tunnel)	Mashhad, Iran
Xia et al. 2026	Surface PM _{2.5} mass and composition under stratospheric sulfate geoengineering	<i>arXiv (preprint)</i>	Chemical transport modelling	Modelled surface PM _{2.5} mass and speciation (dust, sea salt, OC, sulfate, SOA)	Global
Other clinical endpoints					
Chen et al.	Air pollution and sarcopenia: a meta-analysis and network toxicology analysis	<i>Atmos Environ</i>	Metadata only (no abstract)	Air pollution mixture (not extractable)	Not stated
Choi et al.	Time-varying long-term PM _{2.5} and leukaemia incidence from childhood into early adulthood	<i>Environ Res</i>	Nationwide cohort	Annual district PM _{2.5} from a machine-learning ensemble, 226 inland districts	South Korea
Guerrero-Morales et al.	Spatial distribution of long-term ultrafine particle concentrations and systemic lupus erythematosus burden in central Mexico	<i>Environ Toxicol Pharmacol</i>	Ecological / spatial	Satellite-modelled UFP, 1,500-24,000 particles/cm ³	Central Mexico
Kang et al.	Short-term ambient air pollution and acute osteoarthritis morbidity across subgroups	<i>Environ Pollut</i>	Case-crossover	District-level PM ₁₀ , PM _{2.5} , NO ₂ , SO ₂ , O ₃ , CO	South Korea
Lee et al.	Short-term effects and attributable burden of ambient VOC mixtures on paediatric atopic dermatitis and asthma	<i>J Expo Sci Environ Epidemiol</i>	Metadata only (no abstract)	Ambient VOC mixture (not extractable)	Seoul, South Korea
Lei et al.	Hospital admission risk and burden of acute appendicitis due to fine particulate matter pollution	<i>Atmos Environ</i>	Metadata only (no abstract)	PM _{2.5} (not extractable)	China
Park et al.	Short-term fine particulate matter and hepatic decompensation in cirrhosis: a nationwide case-crossover study	<i>Gut Liver</i>	Case-crossover	Daily PM _{2.5} on a 1 km residential grid	South Korea
Poursafa & Jylhava	Aging in the air we breathe: mechanisms and consequences of air pollution exposure for biological aging	<i>Ageing Res Rev</i>	Mechanistic review	PM _{2.5} , PM _{0.1} , TRAP, household combustion (qualitative)	Global
Wang et al.	Early-life PM _{2.5} mass and components and childhood anaemia across China, India and 26 African countries	<i>Ecotoxicol Environ Saf</i>	Cross-sectional (multistage)	MERRA-2 pre-natal/postnatal/cumulative PM _{2.5} + BC, OC, sulfate, dust, sea salt	Multi-country
Wang et al.	Long-term exposure to PM _{2.5} , aging hallmarks and age-related diseases: a nationwide prospective cohort study	<i>Atmos Environ</i>	Metadata only (no abstract)	Long-term PM _{2.5} (not extractable)	Not stated
Wei et al.	Chemical composition of fine particulate matter and all-cause mortality of older lung cancer patients	<i>Environ Res</i>	Cohort	15 PM _{2.5} chemical components, 2-year moving average	USA
Zewdie et al.	Wildfire smoke PM _{2.5} and cancer history prevalence at census-tract level	<i>Cancer Epidemiol</i>	Ecological panel	Daily modelled wildfire-specific smoke PM _{2.5} , 2006-2020	USA
Zhang et al.	PM _{2.5} exposure and the prognosis of severe fever with thrombocytopenia syndrome: an integrated epidemiological and experimental study	<i>Environ Sci Technol</i>	Cohort	Long-term pre-admission ambient PM _{2.5}	Xinyang, China

Provenance and method

This rollup is an **aggregation only**. It pools `state/corpus/*.json` and `state/metrics.csv` over 16 August 2026–22 August 2026 via `PMRW_START/PMRW_END` and **re-queries no literature API**, so every record was screened and deduplicated in the daily issue that first carried it. Per-record provenance, source-query windows, exclusions and known metadata defects are documented in those daily issues and are not restated here. **Daily issues covered: 16, 17, 18, 19, 20, 21 and 22 August 2026 — seven of seven, no gap in entry-date coverage.** The 21 and 22 August issues were built as a backfill on the evening of 22 August after an off-schedule 21 August invocation correctly deferred under the 22:00 rule; the entry windows are contiguous and nothing was skipped. **112 records, 112 unique DOIs** — no duplicate survived the pooling. **Known defects carried into this rollup, stated rather than hidden:** (i) the `ENDPOINT_CANON` gap reported on 15 August is unfixed, so `f2b` understates its “no health endpoint” bar and the caption states the hand-counted figure; (ii) journal names are free text, so preprint share is a lower bound; (iii) **22 records carry no abstract from any checked source and are classified but unassessed** — they are counted in every distribution figure and contribute nothing to the narrative; (iv) the trial registry has not been refreshed since 8 August and no trial section is included in this issue rather than presenting stale counts as current. Effect estimates are transcribed from abstracts and use heterogeneous exposure contrasts — see the forest-plot caption; **no meta-analytic summary is computed and none should be.** Trend and inflection statements are judgements over a seven-day window and are not statistical tests; where a movement is plausibly an artefact of this pipeline’s own harvesting legs, the text says so. Content derived from publisher metadata; abstracts remain the copyright of their respective publishers.